

HYDRAULIC SERVICES

PROPOSED NEW OFFICE BUILDING

STANWELL POWER STATION

SWITCHYARD RD, STANWELL QLD

LOT 44 on SP140243

Rockhampton Regional Council

GENERAL NOTES

- LAYOUTS OF SERVICES SHOWN ON THE ATTACHED DRAWINGS ARE DIAGRAMMATIC ONLY, THEY REPRESENT DESIGN INTENT ONLY IN ACCORDANCE WITH THE RELEVANT CODES AND STANDARDS, AND DOES NOT RELIEVE THE CONTRACTOR OR BUILDER FROM ANY DUE DILIGENCE IN RELATION TO BUILD ABILITY.
- ALL INVERT LEVELS ON THE DRAWINGS ARE RECOMMENDED ONLY AND MUST BE CHECKED ON SITE BEFORE EXCAVATION OR INSTALLATION OF PIPEWORK TO ENSURE CONNECTION TO SUPPLY SERVICES AND CORRECT COVER AND FALL. ADVISE THE SUPERINTENDENT OF ANY APPARENT DISCREPANCIES BEFORE THE COMMENCEMENT OF ANY WORK; CLAIMS FOR REDUNDANT WORK AS A RESULT OF FAILING TO DO SO SHALL NOT BE CONSIDERED VALID.
- ASCERTAIN THE DEPTH, POSITION AND SUITABILITY OF THE SEWER & STORMWATER CONNECTION POINTS PRIOR TO THE COMMENCEMENT OF ANY WORK (PIELAYING & EXCAVATION) AND ENSURE THAT THE SANITARY DRAINAGE & STORMWATER DRAINAGE CAN GRAVITATE TO THE RELEVANT CONNECTION POINTS. THE SUPERINTENDENT SHALL BE ADVISED IMMEDIATELY SHOULD ANY ADJUSTMENT BE REQUIRED TO EXECUTE THE WORK. NO CLAIMS FOR REDUNDANT WORK SHALL BE CONSIDERED DUE TO FAILURE TO COMPLY WITH THIS REQUIREMENT. NO PIPELAYING OR EXCAVATION SHALL BE UNDERTAKEN UNTIL THIS HAS BEEN UNDERTAKEN.
- VISIT THE SITE BEFORE SUBMITTING FINAL TENDER TO ASSESS "ON SITE" CONDITIONS. FAILURE TO DO SO WILL FORFEIT ANY CLAIM FOR NOT BEING AWARE OF CONDITIONS AFFECTING THE TENDER.
- NO WARRANTY IS GIVEN AS TO THE COMPLETENESS OR ACCURACY OF EXISTING SERVICES SHOWN ON THE ATTACHED DRAWINGS.
- OBTAIN MEASUREMENTS AND OTHER NECESSARY INFORMATION AND COORDINATE THE DESIGN AND INSTALLATION IN CONJUNCTION WITH ALL TRADES BEFORE COMMENCING WORK.
- ALL INVERT LEVELS SHOWN ON THE DRAWINGS ARE RECOMMENDATIONS ONLY AND MUST BE CHECKED ON SITE BEFORE EXCAVATION OR INSTALLATION OF PIPEWORK TO ENSURE CONNECTION TO SUPPLY SERVICES AND CORRECT COVER AND FALL CAN BE ACHIEVED. ADVISE THE CONSULTANT OF ANY APPARENT DISCREPANCIES BEFORE THE COMMENCEMENT OF ANY WORK; CLAIMS FOR REDUNDANT WORK AS A RESULT OF FAILING TO DO SO SHALL NOT BE CONSIDERED VALID.
- THE CONTRACTOR SHALL ALLOW FOR ALL NECESSARY DIVERSIONS AND MINOR ADJUSTMENTS TO PIPEWORK AND EQUIPMENT AS MAY BE NECESSARY TO COMPLETE THE WORKS. ALL DIMENSIONING AND DETAILING NECESSARY FOR THE CORRECT PREPARATION OF WORKING DRAWINGS SHALL BE OBTAINED ON SITE AND NOT FROM THE ATTACHED DRAWINGS.
- PRIOR TO COMMENCEMENT OF ANY ASSOCIATED WORKS THE CONTRACTOR SHALL CONFIRM THE PROXIMITY OF EXISTING UTILITY AND SITE SERVICES AND VISIT THE SITE TO BE FULLY AWARE OF SITE CONDITIONS. THE CONTRACTOR IS RESPONSIBLE FOR ANY COSTS ASSOCIATED WITH THE RECTIFICATION OF ANY DAMAGE TO EXISTING UTILITY AND SITE SERVICES, EXISTING STRUCTURES AND THE LIKE CAUSED BY CONTRACTOR NEGLIGENCE.
- THE CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING 'AS CONSTRUCTED DRAWINGS' FOR THE SUBMISSION TO THE LOCAL AUTHORITY AND AS DIRECTED BY THAT AUTHORITY. DRAWINGS MUST BE DRAFTED TO A STANDARD THAT IS THE SAME AS THE ORIGINAL
- IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO ENSURE THAT ALL MATERIALS USED HAVE WATER MARK OR PRODUCT APPROVAL
- SANITARY FIXTURE POINTS SHOWN ON THESE DRAWINGS SHOULD NOT BE DIMENSIONED OR SCALED OFF PIPE SETOUT OR PENETRATIONS SHALL BE DIMENSIONED FROM ARCHITECTURAL DRAWINGS
- BACKFLOW PREVENTION DEVICES SHALL BE INSTALLED WHERE REQUIRED BY THE PLUMBING CODE OF AUSTRALIA (NCC VOLUME 3), AS3500 AND THE LOCAL AUTHORITY HAVING JURISDICTION. DEVICES SHALL BE INSTALLED IN ACCORDANCE WITH AS3500
- THE CONTRACTOR SHALL ALLOW FOR COST OF ALL FEES ASSOCIATED WITH THE PLUMBING WORKS SHOWN ON THESE DRAWINGS PAYABLE TO THE LOCAL AUTHORITY HAVING JURISDICTION
- A SERVICE THAT PENETRATES A BUILDING ELEMENT THAT IS REQUIRED TO HAVE AN FRL MUST COMPLY WITH SECTIONS C4D13, C4D15 OF THE NATIONAL CONSTRUCTION CODE VOLUME 1 (BCA).
- THE CONTRACTOR SHALL ENSURE AND VERIFY WITH BUILDING CERTIFIER THAT FIRE STOP COLLARS HAVE BEEN TESTED IN ACCORDANCE WITH AS1530.4 & AS4072.1 FOR THE PARTICULAR CIRCUMSTANCE OF INSTALLATION.
- A SERVICE LABEL SHALL BE AFFIXED ADJACENT TO EACH SERVICE PENETRATION THAT REQUIRES A F.R.L AND THE LABEL SHALL PROVIDE INFORMATION IN ACCORDANCE WITH AS4072.1
- THE PLUMBING CONTRACTOR SHALL LIAISE WITH STRUCTURAL, MECHANICAL AND ELECTRICAL CONTRACTOR'S BEFORE AND DURING CONSTRUCTION

SANITARY PLUMBING & DRAINAGE

- ALL SANITARY DRAINAGE SHALL BE CONSTRUCTED IN AN APPROVED MATERIAL TO THE REQUIREMENTS OF THE RELEVANT CODES & AUTHORITIES. ALL PIPES SHOWN ARE 1000 HAVING A MINIMUM GRADE OF 1 IN 60 OR AS OTHERWISE SPECIFIED OR DIRECTED.
- ALL PVC FITTINGS USED FOR SANITARY AND PREMISES SEWER SERVICES SHALL COMPLY WITH THE RELEVANT CODES & AUTHORITIES, STANDARDS & SHALL BE JOINTED EITHER BY RUBBER RINGS OR SOLVENT WELDING.
- SUBCONTRACTOR SHALL ALLOW FOR ADDITIONAL EXCAVATION (INCLUDING ROCK) BACK FILL, PIPES, FITTINGS & ALL JUMP UPS TO THE LOCAL AUTHORITY REQUIREMENTS INCLUDING THOSE TO BRANCH DRAINS.
- WHERE DRAINAGE LEVELS PERMIT, ALL BRANCH LINES SHALL BE GRADED &/OR LOWERED TO AVOID PENETRATING FOOTING BEAMS UNLESS SHOWN OTHERWISE. THE CONTRACTOR SHALL ALLOW FOR ADDITIONAL PIPE LENGTHS & WORK INCLUDING CONSTRUCTION OF ADDITIONAL INSPECTION OPENINGS IN THE SYSTEM AS REQUIRED BY LOCAL AUTHORITY.
- CONNECT ALL SANITARY DRAINAGE TO CAPPED SEWER CONNECTIONS AS INDICATED ON THE DRAWINGS UNO.
- PROVIDE INSPECTION OPENINGS IN THE SYSTEM AS REQUIRED BY AS3500.2.2021 SEC 4.7 OR AS REQUIRED BY THE RELEVANT AUTHORITY.
- ALL VENTS TO TERMINATE WITH VENT COWL & ABLE TO ACHIEVE SEPARATION DISTANCES IN ACCORDANCE WITH AS3500.2 2021 FIGURE 6.9.5. ROOF TO BE LEFT WATER TIGHT.

WATER SERVICE NOTES

- INSTALL PIPEWORK IN STRAIGHT LINES & UNIFORM GRADES. ARRANGE & SUPPORT THE PIPEWORK SO THAT IT REMAINS FREE FROM VIBRATION WHILST PERMITTING THERMAL MOVEMENT. KEEP THE NUMBER OF JOINTS TO A MINIMUM. PREVENT DIRECT CONTACT BETWEEN INCOMPATIBLE METALS.
- PROVIDE COMPLIANCE WITH THE NATIONAL CONSTRUCTION CODE 2022 VOLUME 3 SECTION B2D5 AND B2D6 IN RESPECT TO TEMPERATURES OF HEATED DELIVERY OF WATER TO SANITARY FIXTURES.
- STAMPED & TESTED ISOLATING OR LOOSE VALVE STOPCOCKS SHALL BE SUPPLIED & INSTALLED GENERALLY WHERE SHOWN ON THE DRAWINGS & WHEREVER NECESSARY TO PROVIDE CONTROL OF INDIVIDUAL BRANCHES AND/OR GROUPS OF FIXTURES. WHERE APPROPRIATE, RECESS COCKS SHALL BE PROVIDED AS ISOLATORS.
- ALL EXTERNAL HOSE COCKS SHALL BE FITTED WITH A CONTROL COCK LOCATED ON THE RISER & SHALL PROVIDE VACUUM BREAKER FITTING SCREWED TO OUTLET.
- ALL TESTS REQUIRED BY THE AUTHORITIES SHALL BE CARRIED OUT BY THE CONTRACTOR. SUPPLY ALL PLANT & MATERIAL TO CARRY OUT SUCH TESTS & RE-TESTING IF REQUIRED.
- PROVISION SHALL BE MADE TO ENSURE THAT THE MAXIMUM STATIC PRESSURE AT ANY OUTLET, OTHER THAN A FIRE SERVICE OUTLET, WITHIN A BUILDING DOES NOT EXCEED 500 KPA.
- ALL ALLOY BRAZED JOINTS TO BE 5% SILVER SOLDERED.
- NOMINAL PIPE DIAMETERS (DN) SHOWN ON THESE DRAWINGS ARE FOR TYPE B COPPER (U.N.O.), WHERE AN ALTERNATIVE MATERIAL IS PROPOSED, A PIPE MATERIAL WITH AN INTERNAL PIPE DIAMETER EQUIVALENT OR LARGER TO TYPE B COPPER SHALL BE TAKEN FROM AS3500.1 2021 APPENDIX A
- COLD WATER PIPES INSTALLED INGROUND BENEATH CONCRETE SLAB SHALL GENERALLY COMPLY WITH AS3500.1 2021 CLAUSE 5.5.4. PIPE UNDER SLAB FLOOR SHALL BE CONTINUOUS WITHOUT JOINTS. METAL PIPES SHALL BE LAGGED WITH A IMPERMEABLE MATERIAL.
- HOT WATER PIPES INSTALLED INGROUND BENEATH CONCRETE SLAB SHALL GENERALLY COMPLY WITH AS3500.4 CLAUSE 4.6.1.3. PIPE UNDER SLAB FLOOR SHALL BE CONTINUOUS WITHOUT JOINTS AND INSULATED IN ACCORDANCE WITH AS3500.4 2021 CLAUSE 8.2.
- FLOW RESTRICTION DEVICES SHALL BE FITTED WHERE TAP FLOW RATES ARE NOT SELF LIMITING DEVICES SHALL BE USED TO RESTRICT FLOWS TO BASINS, SINKS, LAUNDRY TROUGHS AND SHOWERS TO A MAXIMUM FLOW RATE OF 9L/M OR AS DIRECTED BY THE AUTHORITY HAVING JURISDICTION.
- TAP POINTS SHOWN ON THESE DRAWINGS ARE INDICATIVE ONLY AND DO NOT NECESSARILY REFLECT THE EXACT LOCATION OF SUPPLY TO THE FIXTURE. REFER TO ARCHITECTURAL DRAWINGS AND/OR END CLIENT PC SPECIFICATIONS FOR INSTALLATION REQUIREMENTS AND TO ESTABLISH SUPPLY PROVISIONS
- COLD WATER PIPEWORK SHALL BE IDENTIFIED IN ACCORDANCE WITH AS1345 AND AS3500.1 2021 CLAUSE 5.19
- HOT WATER PIPEWORK SHALL BE IDENTIFIED IN ACCORDANCE WITH AS1345 AND AS3500.4 2021 CLAUSE 4.13.2
- PIPES SHALL BE SEPARATED FROM OTHER SERVICES IN ACCORDANCE WITH AS3500.1 2021 CLAUSE 5.3
- COLD WATER PIPE SYSTEM SHALL BE TESTED AND COMMISSIONED IN ACCORDANCE WITH AS3500.1 2021 SECTION 17
- HOT WATER PIPE SYSTEM SHALL BE TESTED AND COMMISSIONED IN ACCORDANCE WITH AS3500.4 2021 SECTION 9
- HOT WATER PIPE INSULATION SHALL FULLY COMPLY WITH AS3500.4 2021 SECTION 8
- PIPE SIZE LIMITATIONS SHALL BE IN ACCORDANCE WITH AS3500.1.2021 CLAUSE 3.5
- WATER SAVING SHOWERS, BASINS, SINKS & LAUNDRY TROUGHS WITH AAA RATING OR 3 STAR 'WELS' SHALL BE INSTALLED
- MAXIMUM 6-3 (WELS 4 STAR) DUAL FLUSH WC SHALL BE INSTALLED

GUTTER & DOWNPIPE NOTES

- THE DESIGN AND SPECIFICATION OF GUTTERS AND DOWNPIPES ON THESE DRAWINGS ARE TAKEN FROM AS3500.3. CONTRACTOR TO ENSURE GUTTERS AND DOWNPIPES ARE INSTALLED STRICTLY IN ACCORDANCE WITH AS3500.3.
- ROOF DRAINAGE SYSTEM COMPONENTS MADE FROM ALUMINIUM ALLOYS, ALUMINIUM/ZINC ALLOY COATED STEEL, COPPER, COPPER ALLOYS, ZINC COATED STEEL AND ZINC SHALL COMPLY WITH AS/NZS2179.2.
- EAVES GUTTERS
 - STORM EVENT DESIGN FOR GUTTERS ARE AS FOLLOWS:
 - THE IFD DATA USED WAS OBTAINED FROM THE AUSTRALIAN BUREAU OF METEOROLOGY WEBSITE BASED ON THE APPROXIMATE LATITUDE AND LONGITUDE OF THE DEVELOPMENT SITE (LAT: -23.484S, LONG: 150.3257E). THIS DATA INDICATED THE 5 MINUTE, 5% AEP (20 YEAR ARP) RAINFALL INTENSITY APPLICABLE TO THE DEVELOPMENT SITE IS 217mm/hr.
 - AS3500.3 APPENDIX G SHALL BE APPLIED FOR EAVES GUTTERS OVERFLOW MEASURES
 - EXPANSION JOINTS IN EAVES GUTTERS SHALL COMPLY WITH AS3500.3 CLAUSE 4.3.2.
 - SUPPORT SYSTEMS SHALL COMPLY WITH AS3500.3 CLAUSE 4.93
 - GUTTER DOWNPIPES SHALL BE DESIGNED AND INSTALLED IN ACCORDANCE WITH AS3500.3 2021 SECTION 3.5.3, AND 4.5.6
 - VALLEY GUTTERS
 - ROOF SLOPES OF NOT LESS THAN 12.5°
 - VALLEY CATCHMENTS SHALL NOT EXCEED 20m²
 - METHOD OF INSTALLATION SHALL COMPLY WITH AS3500.3 2021 CLAUSE 4.5.5. DESIGN SHALL BE IN ACCORDANCE WITH AS3500.3 2021 SECTION 3.6
 - SUPPORT SYSTEMS SHALL COMPLY WITH AS3500.3 CLAUSE 4.9.5
- DOWNPIPES
 - METHOD OF INSTALLATION SHALL COMPLY WITH AS3500.3 CLAUSE 4.5.6. DESIGN SHALL BE IN ACCORDANCE WITH AS3500.3 2021 SECTION 3.5.3, 3.7.8, 4.5.6 AND 4.9.6
 - DOWNPIPES AND HORIZONTAL STORMWATER DRAINS INSTALLED ABOVE GROUND SHALL BE FIXED AND SUPPORTED IN ACCORDANCE WITH AS3500.3 AND MANUFACTURERS INSTRUCTIONS
 - ARRANGE PIPING TO ACCOMMODATE THERMAL EXPANSION. PROVIDE PROPRIETARY EXPANSION JOINT IN PLASTIC PIPES WHERE PIPE FLEXIBILITY DOES NOT ALLOW MOVEMENT. MAKE SURE MOVEMENT DOES NOT STRAIN BRANCH CONNECTIONS.

SOIL CLASSIFICATION TABLE			
AS2870 CLASSIFICATION	DIFFERENTIAL MOVEMENT	RECOMMENDED MINIMUM GRADE	SOIL FOUNDATION
M	20-40mm	1:60	MODERATELY REACTIVE CLAY SITES, WHICH MAY EXPERIENCE MODERATE GROUND MOVEMENT FROM MOISTURE CHANGE. NO PIPE ARTICULATION REQUIRED FOR THIS SITE

NOTE: TABLE SHOULD BE USED AS A GUIDE ONLY, REFER TO GEOTECHNICAL REPORT FOR DETAILS



1 Locality Plan

NOT TO SCALE

FIRE HYDRANT SERVICE NOTES

- THE WORK COVERED BY THIS SECTION SHALL INCLUDES THE INSTALLATION OF ALL PIPING, FITTINGS, VALVES, HYDRANT VALVES TO AS3500.1, AS2941.
- THIS SECTION OF WORK IS FOR THE SUPPLY & INSTALLATION OF THE FIRE HYDRANT SERVICE FROM THE EXISTING MAINS TO ALL FIRE HYDRANTS SHOWN ON THE DRAWINGS.
- FIRE HYDRANTS SHALL BE INSTALLED IN ACCORDANCE WITH AS2419:2021 & BE STAMPED APPROVED BY THE RELEVANT AUTHORITY. HYDRANT LANDING VALVES SHALL BE COMPATIBLE WITH THE QLD ROUND THREAD PATTERN.
- PROVIDE SIGNAGE TO FIRE HYDRANTS INSTALLATION IN ACCORDANCE WITH AS2419 OR AS INSTRUCTED BY THE AUTHORITY.
- A NON CERTIFIED MATERIAL USED FOR THE SOLE PURPOSE OF A FIRE SERVICE INSTALLATION SHALL BE INSTALLED DOWNSTREAM OF A TESTABLE BACKFLOW PREVENTION DEVICE AND/OR AS DIRECTED BY THE AUTHORITY HAVING JURISDICTION.
- EXTERNAL HYDRANTS SHALL BE POSITIONED NOT LESS THAN 10M FROM ANY PART OF THE BUILDING UNLESS PROTECTED BY A CONSTRUCTION HAVING A FRL OF 90/90/90 EXTENDING AT LEAST 2M EITHER SIDE OF THE HYDRANT AND 3M ABOVE FINISHED GROUND LEVEL. (UNLESS INSTRUCTED OTHERWISE)
- HYDRANTS SHALL BE POSITIONED MINIMUM OF 10M AWAY FROM ANY HIGH VOLTAGE MAIN ELECTRICAL EQUIPMENT, LPG AND ANY OTHER COMBUSTIBLE STORAGE.
- THE HYDRANT VALVE OUTLET SHALL BE FITTED WITH AN APPROVED PVC/METAL CAP AND STAINLESS STEEL CHAIN OR WIRE. PROVIDE AN APPROVED METAL CAP ON VALVE TURNING WHEELS WITH 003 LOCKS ON ALL HYDRANT VALVES.
- PROTECTIVE BOLLARDS SHALL BE PROVIDED AROUND THE FOLLOWING FH INSTALLATION LOCATIONS:
 - ALL FH THAT ARE WITHIN 1.5M OF A VEHICLE ACCESSIBLE PATH – MINIMUM 2 BOLLARDS.
 - ALL FH WHERE A VEHICLE COULD APPROACH/IMPINGE FROM MULTIPLE DIRECTIONS MINIMUM 4 BOLLARDS.
 - ALL FH IN GRASSED AREAS AND/OR IN OPEN AREAS (OUT IN THE MIDDLE) – MINIMUM 4 BOLLARDS.THE PROTECTIVE BOLLARDS SHALL BE INSTALLED TO APPLICABLE CODES & STANDARDS AND AS A MINIMUM:
 - BE MADE FROM DN150 Ø HEAVY DUTY GALVANISED STEEL PIPES.
 - BE PAINTED WITH HIGH-VISIBILITY YELLOW POWDER COATED FINISH.
 - BE MINIMUM 1200 MM IN HEIGHT FROM FINISHED GROUND LEVEL.
 - BE CORE-FILLED WITH CONCRETE TO IMPROVE RESISTANCE/RIGIDITY/STABILITY.
 - SHALL HAVE A MINIMUM 480MM Ø (OR SQUARE) x 600MM DEEP CONCRETE BASE AND BE SECURED BY 4 x 300MM LONG M16 BOLTS ON A MINIMUM 320MM Ø (OR SQUARE) x 10MM THICK BASEPLATE.
 - THE CONCRETE BASE/BASEPLATE SHALL BE SUITABLY INSTALLED / CHAMFERED TO PREVENT TRIP HAZARDS.
- HYDRANT SYSTEM SHALL BE COMMISSIONED IN ACCORDANCE WITH AS2419.1:2021 APPENDIX S
- HYDRANT SYSTEM SHALL BE SUPPORTED IN ACCORDANCE WITH AS2419.1 2021 SECTION 10.
- HYDRANT SYSTEM SHALL BE IDENTIFIED IN ACCORDANCE WITH AS2419.1 2021 SECTION 9.7
- INSTALL PIPE SUPPORTS (THRUST BLOCKS) IN ACCORDANCE WITH AS3500.1

STORMWATER DRAINAGE NOTES

- ALL PIPES SHALL BE BEDDED IN ACCORDANCE WITH AS3500.3 (UNO).
- ALL PIPES SHALL BE LAID IN SUCH A MANNER THAT THEIR BARRELS BEAR FIRMLY & EVENLY ON THE BEDDING MATERIAL. THE BACKFILL SHALL CONSIST OF APPROVED FILLING FREE FROM ROCK OR HARD SUBSTANCES. PIPES ARE TO BE BACK FILLED TO AT LEAST 2/3 OF THEIR DIAMETER WITH COMPACTED SAND OR CRUSHER RUN. ALL BACKFILL SHALL BE PLACED SO AS NOT TO DAMAGE THE PIPE & SHALL BE ADDED IN LAYERS OF NOT MORE THAN 300mm & BE THOROUGHLY COMPACTED. ALL PIPES LAID UNDER FLOOR SLABS OR PAVING SHALL BE BACKFILLED WITH COMPACTED SAND. REPLACE BATTER WITH NATURAL MATERIAL COMPACTED TO 98 % STANDARD MAXIMUM DRY DENSITY.
- CONNECT STORMWATER DRAINAGE TO THE CIVIL STORMWATER DRAINAGE WHERE SHOWN ON THE DRAWINGS, ALL WORKS TO COMPLY WITH THE REQUIREMENTS OF THE RELEVANT AUTHORITIES. CONTRACTOR SHALL MAKE ARRANGEMENTS FOR INSPECTIONS IF REQUIRED BY THE AUTHORITY (COUNCIL), PRIOR TO BACKFILLING.
- MINIMUM GRADE TO STORMWATER LINES SHALL BE 1% (U.N.O.).
- ALL PIPEWORK JUNCTIONS AND TAPERS UP TO & INCLUDING 225 DIA SHALL BE VIA MANUFACTURED MADE FITTINGS
- LAY PIPELINES TO UNIFORM GRADIENTS, STRAIGHT BETWEEN REQUIRED CHANGES OF DIRECTION, PROPERLY SUPPORTED, WITH WATERTIGHT JOINTS AND WITH SPIGOT ENDS POINTED IN THE DIRECTION OF FLOW. PROVIDE THE NECESSARY FITTINGS AND ACCESSORIES, INCLUDING JUNCTIONS, BRANCHES, INSPECTION AND CLEANING OPENINGS, EXPANSION JOINTS, AND THE LIKE.
- CONTRACTOR TO SUPPLY & INSTALL ALL FITTINGS & SPECIALS INCLUDING VARIOUS PIPE ADAPTERS.
- REFER CIVIL STRUCTURAL DRAWINGS FOR INFORMATION REGARDING SLABS, KERBS, RETAINING WALLS, VEHICLE LAYBACKS, ROAD LEVEL AND SITE LEVELS.
- MINIMUM CLEAR COVER TO PIPELINES SHALL BE IN ACCORDANCE WITH AS3500.3.

Issued for
TENDER ONLY

NOT FOR
CONSTRUCTION

DRAWING SHOULD BE
READ IN COLOUR PRINT

01	ISSUED FOR TENDER	24.02.2026
P1	ISSUED FOR REVIEW	12.01.2026

ISSUE:	DESCRIPTION:	DATE:
	HYDRO-TECH DESIGNS PTY LTD HYDRAULIC AND FIRE PROTECTION CONSULTANTS Tel: 0438 433 544 Email: rjch@bigpond.net.au	

CLIENT:



BUILDING DESIGNER



PROJECT:
NEW OFFICE BUILDING

STANWELL POWER STATION
SWITCHYARD ROAD, STANWELL

TITLE:
HYDRAULIC SERVICES
COVER SHEET
NOTES
LOCALITY PLAN

Datum:	Status:	
TENDER		
Scale:	Date:	
AS NOTED	19.11.2025	
Project Number.	Sheet	Issue.
HS-0690-b	H.01.01	01

SEPARATION OF SERVICES	
ABOVE GROUND WATER SERVICE	
FROM ELECTRICAL CONDUIT	MIN. 25mm
FROM ELECTRICAL WIRE OR CABLE	MIN. 25mm
FROM CONSUMER GAS PIPES	MIN. 25mm
UNDERGROUND WATER SERVICE	
FROM ELECTRICAL SUPPLY CABLE	MIN. 100mm FOR WATER SERVICE <DN65 PROVIDED THE ELECTRICAL CABLE IS INDICATED ALONG ITS LENGTH WITH ORANGE MARKER TAPE COMPLYING WITH AS2648.1 AND IS MECHANICALLY PROTECTED.
	300mm FOR WATER SERVICE >DN65 PROVIDED THE ELECTRICAL CABLE IS INDICATED ALONG ITS LENGTH WITH ORANGE MARKER TAPE COMPLYING WITH AS2648.1 AND IS MECHANICALLY PROTECTED.
FROM CONSUMER GAS PIPES	MIN. 600mm WHERE ELECTRICAL SUPPLY CABLE IS NEITHER NOR MECHANICALLY PROTECTED IN ACCORDANCE WITH THE ABOVE
	MIN. 100mm FOR WATER SERVICE <DN65 PROVIDED THE CONSUMER GAS PIPE IS INDICATED ALONG ITS LENGTH WITH MARKER TAPE COMPLYING WITH THE REQUIREMENTS OF AS2648.1 AND LAID 150mm ABOVE THE INSTALLED PIPE.
	300mm FOR WATER SERVICE >DN65 PROVIDED THE CONSUMER GAS PIPE IS INDICATED ALONG ITS LENGTH WITH MARKER TAPE COMPLYING WITH THE REQUIREMENTS OF AS2648.1 AND LAID 150mm ABOVE THE INSTALLED PIPE.
	MIN. 600mm WHERE CONSUMER GAS PIPE IS NEITHER NOR MECHANICALLY PROTECTED IN ACCORDANCE WITH THE ABOVE
FROM COMMUNICATION CABLE	100mm MINIMUM
FROM UNDERGROUND DRAINS	WATER SERVICE PIPES SHALL NOT BE LAID IN OR THROUGH ANY DRAIN
FROM NON-DRINKING WATER PIPES	ABOVE GROUND INSTALLATIONS OF NON-DRINKING WATER SERVICES SHALL NOT BE INSTALLED WITHIN 100mm OF ANY PARALLEL DRINKING WATER SUPPLY
	BELOW GROUND INSTALLATIONS OF NON-DRINKING WATER SERVICES SHALL NOT BE INSTALLED WITHIN 300mm OF ANY PARALLEL DRINKING WATER SUPPLY
CROSSOVER OF UNDERGROUND SERVICES	
ANY UNDERGROUND WATER SERVICE CROSSING ANOTHER SERVICE OR ANY UNDERGROUND SERVICE CROSSING A WATER SERVICE	CROSS AT AN ANGLE OF NOT LESS THAN 45°
	HAVE VERTICAL SEPARATION OF NOT LESS THAN 100mm
	BE MARKED ALONG ITS LENGTH FOR 1m EITHER SIDE OF THE CENTRE-LINE OF THE SERVICE WITH MARKER TAPE COMPLYING WITH AS2648.1 AND LAID 150mm ABOVE THE INSTALLED SERVICE

SEPARATION OF SERVICES	
ABOVE GROUND SANITARY DRAINAGE	
FROM ELECTRICAL CONDUIT	MIN. 100mm
FROM ELECTRICAL WIRE OR CABLE	MIN. 100mm
FROM CONSUMER GAS PIPES	MIN. 100mm
FROM WATER PIPES	MIN. 100mm

BELOW GROUND SANITARY DRAINAGE	
FROM ELECTRICAL SUPPLY CABLE	MIN. 100mm PROVIDED THE ELECTRICAL CABLE IS INDICATED ALONG ITS LENGTH WITH ORANGE MARKER TAPE COMPLYING WITH AS2648.1 AND IS MECHANICALLY PROTECTED
	MIN. 600mm WHERE ELECTRICAL CABLE IS NEITHER INDICATED OR PROTECTED
FROM CONSUMER GAS PIPE	MIN. 100mm PROVIDED THE CONSUMER GAS PIPE IS INDICATED ALONG ITS LENGTH WITH MARKER TAPE COMPLYING WITH THE REQUIREMENTS OF AS2648.1 AND IS LAID 150mm ABOVE THE INSTALLED PIPE AND IS MECHANICALLY PROTECTED.
600mm WHERE CONSUMER GAS PIPE IS NEITHER NOR MECHANICALLY PROTECTED	
FROM COMMUNICATION CABLE	100mm MINIMUM

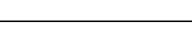
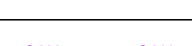
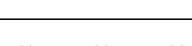
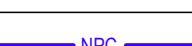
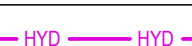
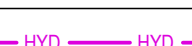
CROSSOVER OF UNDERGROUND SERVICES	
ANY UNDERGROUND DRAIN CROSSING ANOTHER SERVICE OR ANY UNDERGROUND SERVICE CROSSING A DRAIN	CROSS AT AN ANGLE OF NOT LESS THAN 45°
	HAVE VERTICAL SEPARATION OF NOT LESS THAN 100mm
	BE MARKED ALONG ITS LENGTH FOR 1m EITHER SIDE OF THE CENTRE-LINE OF THE SERVICE WITH MARKER TAPE COMPLYING WITH AS2648.1 AND LAID 150mm ABOVE THE INSTALLED SERVICE

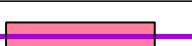
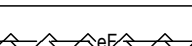
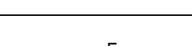
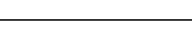
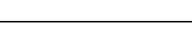
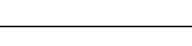
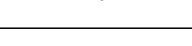
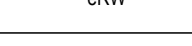
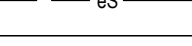
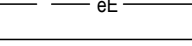
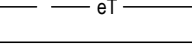
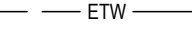
LEGEND	
	CONNECTOR (UNION)
	AIR ADMITTANCE VALVE
	CAPPED END
	BALL VALVE
	SINGLE CHECK VALVE
	DUAL CHECK VALVE
	DOUBLE CHECK VALVE (TESTABLE)
	REDUCED PRESSURE ZONE DEVICE (TESTABLE)
	DUAL PILLAR HYDRANT (RISE)
	36m FIRE HOSE REEL
	WATER METER ASSEMBLY
	OVERFLOW RELIEF GULLY/DISCONNECTOR GULLY
	END LINE
	OUTLET
	EXPANSION JOINT
	FLOOR WASTE GULLY
	LOCKED ISOLATION VALVE
	ISOLATION VALVE IN PATH BOX
	GATE VALVE
	PIPE DROP
	S AND/OR P TRAP
	PIPE DROP
	PUMP
	REDUCER
	PRESSURE LIMITING/REDUCTION VALVE
	HOSE TAP - RISER
	HOSE TAP - DROPPER
	TEMPERING VALVE (50° MAX.)
	THERMOSTATIC MIXING VALVE (45° MAX)
	TAP POINTS (HOT)
	TAP POINTS (COLD)
	TAP POINTS (TEPID)
	HYDRANT/FIRE HOSE REEL HOSE LENGTH LABEL
	PROPOSED LEVEL
	EXISTING LEVEL
	RISER SERVICE TYPE
	SIZE
	ITEM NUMBER
	RISER SERVICE TYPE
	SIZE
	DROPPER

MINIMUM PIPE GRADES (SANITARY DRAINAGE)	
Nominal Size (DN)	Minimum Grade (%)
40-65	2.50
80	1.65
100	1.65*
* except for drains from septic tanks, sewage treatment plants and unvented discharge pipes from tundishes, which may have a grade of 1.00%	

EQUIVALENT PIPE DIAMETERS (DN)						
DN	COPPER (TYPE B)	PE (SDR 11)	PVC-M - PN16	PE-Xa	PP-R (SDR 9)	GAL STEEL (MEDIUM)
15	15 (10.88mm ID)	16 (12.7mm ID)	-	16 (10.8mm ID)	-	15 (16.1mm ID)-
18	18 (13.84mm ID)	20 (16mm ID)	-	20 (13.6mm ID)	-	-
20	20 (17.01mm ID)	25 (20.2mm ID)	-	25 (18.0mm ID)	-	20 (21.7mm ID)
25	25 (22.96mm ID)	32 (26mm ID)	-	32 (23.2mm ID)	32 (24.87mm ID)	25 (27.3mm ID)
32	32 (29.31mm ID)	40 (32mm ID)	-	40 (29.9mm ID)	40 (31mm ID)	32 (36mm ID)
40	40 (35.66mm ID)	50 (40.4mm ID)	-	50 (36.2mm ID)	50 (38.8mm ID)	40 (41.9mm ID)
50	50 (48.36mm ID)	63 (51mm ID)	-	63 (45.8mm ID)	63 (48.81mm ID)	50 (53.1mm ID)
65	65 (61.06mm ID)	75 (61mm ID)	-	--	90 (69.8mm ID)	65 (68.9mm ID)
80	80 (72.94mm ID)	90 (73mm ID)	-	-	110 (85.4mm ID)	80 (80.9mm ID)
100	100 (98.34mm ID)	125 (101.5mm ID)	100 (110.3mm ID)	-	160 (124.2mm ID)	100 (105.3mm ID)

ABBREVIATIONS	
AP	ACCESS PANEL
AAV	AIR ADMITTANCE VALVE
BTH	BATH
BM	BAINE MARIE
BN	BASIN
BWU	BOILING WATER UNIT
CV	CHECK VALVE
CWM	CLOTHES WASHING MACHINE
CS	CLEANERS SINK
CW	COLD WATER
DCV	DOUBLE CHECK VALVE
DP	DOWNPIPE
DG	DISCONNECTOR GULLY
DF	DRINKING FOUNTAIN
DPH	DUAL PILLAR HYDRANT
DWM	DISH WASHING MACHINE
EJ	EXPANSION JOINT
EPD	ELEVATED PIPE DRAIN
FF	FROG FLAP
FFL	FINISHED FLOOR LEVEL
FHR	FIRE HOSE REEL
FU	FIXTURE UNIT
FGL	FINISHED GROUND LEVEL
FSL	FINISHED SURFACE LEVEL
FS	FIRE SERVICE
FW	FLOOR WASTE
FWG	FLOOR WASTE GULLY
GIT	GREASE INTERCEPTOR TRAP
GWM	GLASS WASHING MACHINE
HC	HOSE COCK
HCB	HOUSE CONNECTION BRANCH
HDGS	HOT DIPPED GALVANISED STEEL
HYD	HYDRANT (SPRING)
HWU	HOT WATER UNIT
HWF	HOT WATER FLOW
HWR	HOT WATER RETURN
HW	HOT WATER
HTD	HYDRANT TEST DRAIN
IC	INSPECTION CHAMBER
IJ	INSPECTION JUNCTION
IL	INVERT LEVEL
IM	ICE MACHINE
IO	INSPECTION OPENING
IOS	INSPECTION OPENING TO SURFACE
IWD	INDIVIDUAL WASTE DROPPER
NRV	NON RETURN VALVE
ORG	OVERFLOW RELIEF GULLY
PBO	PLANTER BOX OUTLET
PE-HD	HIGH DENSITY POLYETHYLENE
PG	PIPE GUARD
RHS	RECTANGULAR HOLLOW SECTION
RL	RELATIVE LEVEL
RPD	REAL PROPERTY DESCRIPTION
RRJ	RUBBER RING JOINT
RWH	RAINWATER HEAD
RV	RELIEF VALVE
SK	SINK
SHR	SHOWER
SL	SURFACE LEVEL
STY	SAFETRAY
ST	STACK
SW	STORMWATER
SWJ	SOLVENT WELDED JOINT
TBM	TEMPORARY BENCH MARK
TD	TUNDISH
TMV	THERMOSTATIC MIXING VALVE
TPV	TEMPERING VALVE
TTD	TRAPPED TUNDISH
TG	TEST GATE
TR	TROUGH
TW	TRADE WASTE
U	URINAL
UNO	UNLESS NOTED OTHERWISE
WC	WATER CLOSET

MATERIALS - STORMWATER			
uPVC PIPES AND FITTINGS UP TO 300mm DIAMETER CONSTRUCTED TO AS1254, AS2032 & AS1273.			
ABOVE GROUND			
LINETYPE		SIZE	MATERIAL
	STORMWATER	DN100	DWV PIPE & FITTINGS SN6, SOLVENT WELD JOINT
	STORMWATER	DN150-DN300	DWV PIPE & FITTINGS SN4 OR SN8, SOLVENT WELD JOINT
BELOW GROUND			
LINETYPE		SIZE	MATERIAL
	STORMWATER	DN100	DWV PIPE & FITTINGS SN6, SOLVENT WELD JOINT
	STORMWATER	DN150-DN300	DWV PIPE & FITTINGS SN4 OR SN8, SOLVENT WELD JOINT
MATERIALS - SANITARY PLUMBING, DRAINAGE & VENTS			
ABOVE GROUND			
LINETYPE		SIZE	MATERIAL
	SANITARY PLUMBING	DN40-DN80	DWV PIPE & FITTINGS SH, SOLVENT WELD JOINT
	SANITARY PLUMBING	DN100	DWV PIPE & FITTINGS SN6 OR SN10, SOLVENT WELD JOINT
	VENT PIPE	AS NOTED	DWV PIPE & FITTINGS SH, SOLVENT WELD JOINT DWV PIPE & FITTINGS SN6 OR SN10, SOLVENT WELD JOINT
BELOW GROUND			
LINETYPE		SIZE	MATERIAL
	SANITARY DRAINAGE	DN40-DN80	DWV PIPE & FITTINGS SH, SOLVENT WELD JOINT
	SANITARY DRAINAGE	DN100	DWV PIPE & FITTINGS SN6 OR SN10, SOLVENT WELD JOINT
	VENT PIPE	AS NOTED	DWV PIPE & FITTINGS SH, SOLVENT WELD JOINT DWV PIPE & FITTINGS SN6 OR SN10, SOLVENT WELD JOINT
MATERIALS - WATER SERVICES			
ABOVE GROUND - COLD			
LINETYPE		SIZE	MATERIAL
	POTABLE WATER	DN16-DN150	TYPE B COPPER WITH SILVER BRAZED OR APPROVED CRIMPED JOINTS
	POTABLE WATER	DN16-DN63	REHAU RAUTITAN PE-XA - PLATINUM
	NON POTABLE	DN16-DN63	REHAU RAUTITAN PE-XA - PLATINUM - TYPE B COPPER WITH SILVER BRAZED OR APPROVED CRIMPED JOINTS
BELOW GROUND - COLD			
LINETYPE		SIZE	MATERIAL
	POTABLE WATER	DN16-DN150	PE100 SERIES PRESSURE PIPE (BLUELINE) CLASS 12.5, COMPRESSION JOINTS
	POTABLE WATER	DN16-DN63	REHAU RAUTITAN PE-XA - PLATINUM
	NON POTABLE	DN16-DN150	PE100 SERIES PRESSURE PIPE (BLUELINE) CLASS 12.5, COMPRESSION JOINTS
ABOVE GROUND - HOT			
LINETYPE		SIZE	MATERIAL
	POTABLE WATER (60°)	DN16-DN150	TYPE B COPPER WITH SILVER BRAZED OR APPROVED CRIMPED JOINTS
	POTABLE WATER (60°)	DN16-DN63	REHAU RAUTITAN PE-XA - PLATINUM
	POTABLE WATER (45°-50°)	DN16-DN63	REHAU RAUTITAN PE-XA - PLATINUM
MATERIALS - FIRE SERVICES			
ABOVE GROUND			
LINETYPE		SIZE	MATERIAL
	HYDRANTS	DN100-DN150	LIGHT WALL HOT DIPPED GAL STEEL PIPE, ROLL GROOVED JOINTS
BELOW GROUND			
LINETYPE		SIZE	MATERIAL
	HYDRANTS	DN100-DN150	PVC-M SERIES 2 (BLUE RHINO) PN16 , RUBBER RING JOINTS
	HYDRANTS	DN100-DN150	ELECTROFUSION FITTINGS, BUTT WELD JOINTS

LINETYPES	
	COUNCIL WATER
	FIRE SERVICE DIAGRAMMATICS
	SANITARY DIAGRAMMATICS
	PIPE THROUGH BEAM/FOOTING
	REDUNDANT SERVICE
	FIRE HYDRANT HOSE TRAVEL
	FIRE HOSE REEL HOSE TRAVEL
	EXISTING FIRE SERVICE
	EXISTING SANITARY DRAIN
	EXISTING STORMWATER
	EXISTING WATER SERVICE
	EXISTING RAW WATER
	EXISTING SEWER
	EXISTING ELECTRICAL MAIN
	EXISTING TELECOMMUNICATION
	EXISTING TRADE WASTE
	CIVIL STORMWATER

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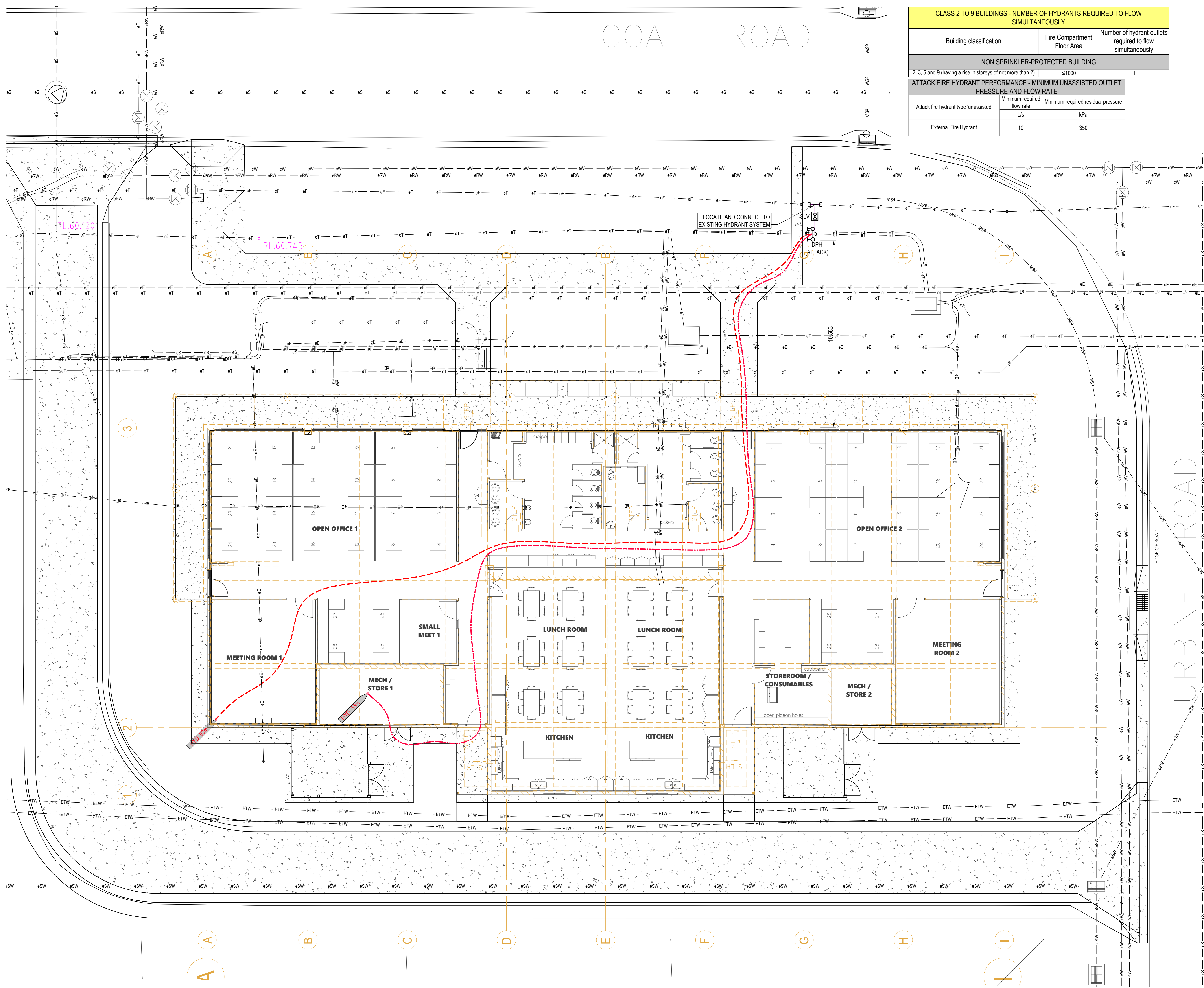
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P1	ISSUED FOR REVIEW	12.01.2026
ISSUE:	DESCRIPTION:	DATE:
	HYDRO-TECH DESIGNS PTY LTD HYDRAULIC AND FIRE PROTECTION CONSULTANTS Tel: 0438 433 544 Email: rjch@bigpond.net.au	



PROJECT:	NEW OFFICE BUILDING
STANWELL POWER STATION SWITCHYARD ROAD, STANWELL	
TITLE:	HYDRAULIC SERVICES SPECIFICATION NOTES SCHEDULES

Datum:	Status:	
TENDER		
Scale:	Date:	
	19.11.2025	
Project Number.	Sheet	Issue.
HS-0690-b	H.01.02	01



CLASS 2 TO 9 BUILDINGS - NUMBER OF HYDRANTS REQUIRED TO FLOW SIMULTANEOUSLY		
Building classification	Fire Compartment Floor Area	Number of hydrant outlets required to flow simultaneously
NON SPRINKLER-PROTECTED BUILDING		
2, 3, 5 and 9 (having a rise in storeys of not more than 2)	≤1000	1
ATTACK FIRE HYDRANT PERFORMANCE - MINIMUM UNASSISTED OUTLET PRESSURE AND FLOW RATE		
Attack fire hydrant type 'unassisted'	Minimum required flow rate	Minimum required residual pressure
	L/s	kPa
External Fire Hydrant	10	350

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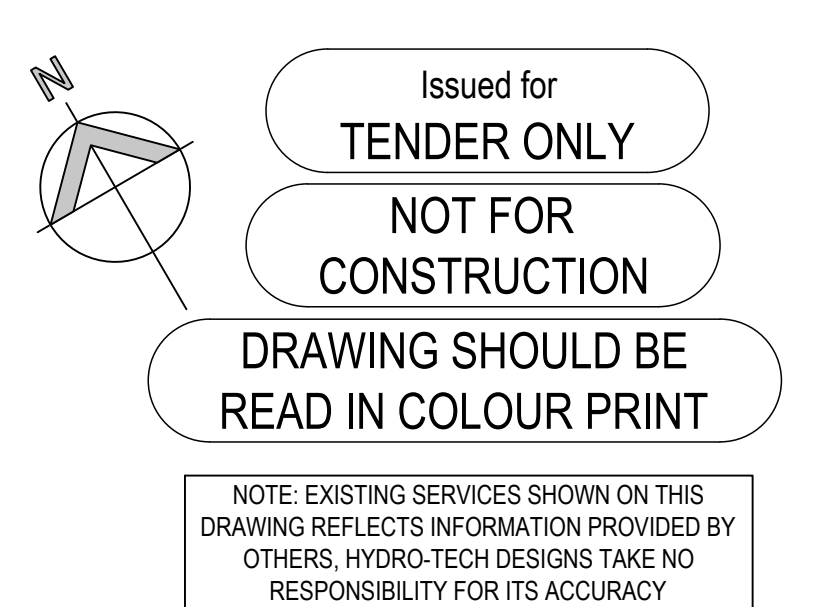
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PROJECT:
NEW OFFICE BUILDING

**STANWELL POWER STATION
SWITCHYARD ROAD, STANWELL**

TITLE:
**HYDRAULIC SERVICES
SITE AND GROUND FLOOR
FIRE HYDRANT AND FIRE HOSEREEL
HOSE REACH PLAN**

Datum:		Status:	
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Project Number.		Sheet	Issue.
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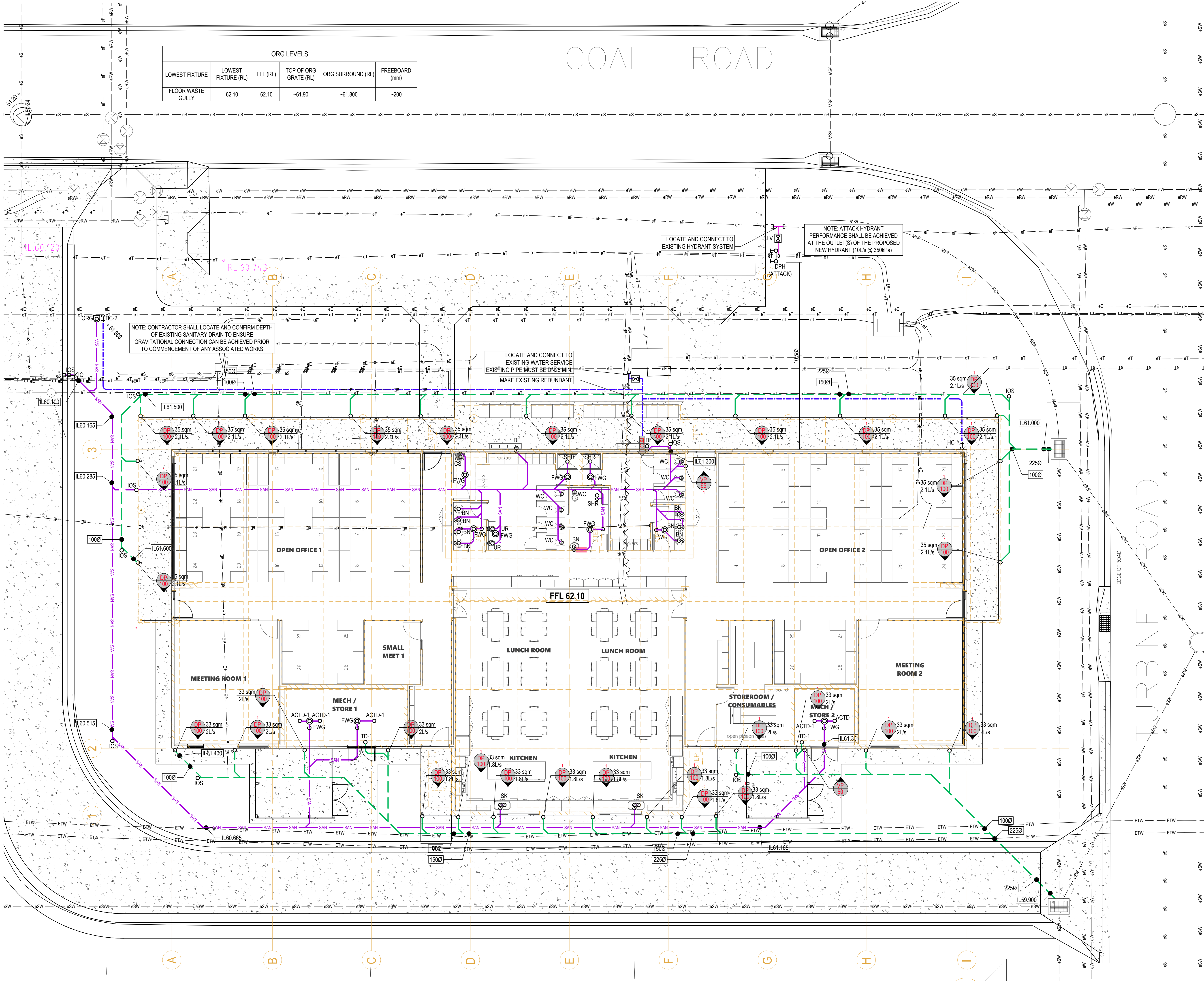
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STANWELL POWER STATION
SWITCHYARD ROAD, STANWELL

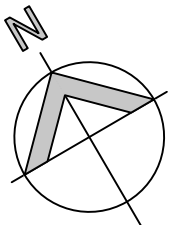
TITLE:
HYDRAULIC SERVICES
DEMOLITION PLAN

Datum:		Status:	
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ORG LEVELS					
LOWEST FIXTURE	LOWEST FIXTURE (RL)	FFL (RL)	TOP OF ORG GRATE (RL)	ORG SURROUND (RL)	FREEBOARD (mm)
FLOOR WASTE GULLY	62.10	62.10	-61.90	-61.800	-200

COAL ROAD



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SANITARY DRAIN SCHEDULE	
FIXTURE TYPE	SIZE
SINK TO DRAIN	50 TRAPPED
WC TO DRAIN	100
BASIN TO DRAIN	40 TRAPPED
DF TO DRAIN	50 TRAPPED
URINAL TO DRAIN	40 TRAPPED, PIPE IN WALL
BASIN TO FWG	40 TRAPPED, PIPE IN WALL
ITWD TO FWG	40
SHR TO FWG	50 (80 or 100 PALAZZI TRAP IF >2.5 FROM FWG)
SHR TO DRAIN	100x65 PALAZZI GULLY
CS TO FWG	50 TRAPPED

NOTE: EXISTING SERVICES SHOWN ON THIS DRAWING REFLECTS INFORMATION PROVIDED BY OTHERS. HYDRO-TECH DESIGNS TAKE NO RESPONSIBILITY FOR ITS ACCURACY

NOTE: THE LOCATIONS OF TUNDISHES FOR THE COLLECTION OF A/C CONDENSATE SHOWN ON THESE DRAWINGS ARE INDICATIVE ONLY AND SHALL BE PROPERLY COORDINATION WITH THE MECHANICAL SERVICES CONTRACTORS

LEGEND: DRAINAGE	
TD-1	100 GRATED DRAIN FOR THE COLLECTION OF RPZD WASTE
ACTD-1	100 x 50 LIT FOR THE COLLECTION OF A/C CONDENSATE WASTE
LEGEND: WATER	
HC-1	HOSE TAP ON WALL/POST WITH ISOLATION VALVE UPSTREAM
HC-4	HOSE TAP ON POST SPECIFICALLY TO MAIN TRAP SEAL IN ORG

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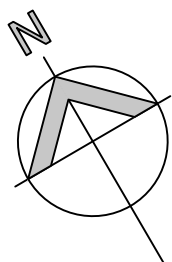


PROJECT:
NEW OFFICE BUILDING

STANWELL POWER STATION
SWITCHYARD ROAD, STANWELL

TITLE:
HYDRAULIC SERVICES
SITE AND GROUND FLOOR
DRAINAGE, STORMWATER, WATER
& FIRE SERVICES

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LEGEND: WATER	
HWU-1	ELECTRIC INSTANTANEOUS HOT WATER UNIT EQUAL TO ZIP DEX 27948 (60") 27kW 3Ph 400V 3 phase. Fully electronically controlled instantaneous water heater for single or multipoint applications. Servotronic control automatically regulates the water flow rate for optimum performance. 600C model is suitable for installations which do not require temperature control or for use with a TMV. The integrated real glass touch display allows individual user profile settings and temperature settings.
BCHC-1	UNDERBENCH CHILLED/BOILING/HOT & COLD WATER UNIT EQUAL TO ZIP HYDROTAP G5 BCH440 CLASSIC ALL IN ONE 1.97kW 220-240V 50/60Hz 1 x 10A GPO. INSTALL TO MANUFACTURERS Zip HydroTap Classic All-in-One Boiling, Chilled, Filtered water plus Hot and Cold washing up water from a single tap (cold water supply required). The system features a single under bench command centre including full colour interactive touch screen display with pin code protection and customisable settings including 2 boiling water safety modes, 3 energy saving modes including ON/OFF timers and sensor activated 'sleep when it's dark', 'Quiet' mode with automated intelligent fan speed adjustment, StenTouch® antimicrobial protection impregnated into key water paths for enhanced hygiene, 100% water efficient air cooled technology, and 0.2 micron water filtration. Easy to install integrated ventilation tray included.
TMV-1	THERMOSTATIC MIXING VALVE IN STAINLESS STEEL CABINET EQUAL TO AQUABLEND 1000 MODEL ATMS710R-350 (3 PIPE CABINET) 15mm INLET & 20mm OUTLET FLOW RATE 39L/m @ 300kPa PRESSURE LOSS
HC-3	HOSE TAP ON WALL/POST WITH RPZD UPSTREAM AND 'DO NOT DRINK' SIGN ADJACENT
HC-4	SHROUDED C/P HOSE TAP ON AT LOW LEVEL ADJACENT TO URINAL SPECIFICALLY TO MAINTAIN WATER SEAL IN FWG

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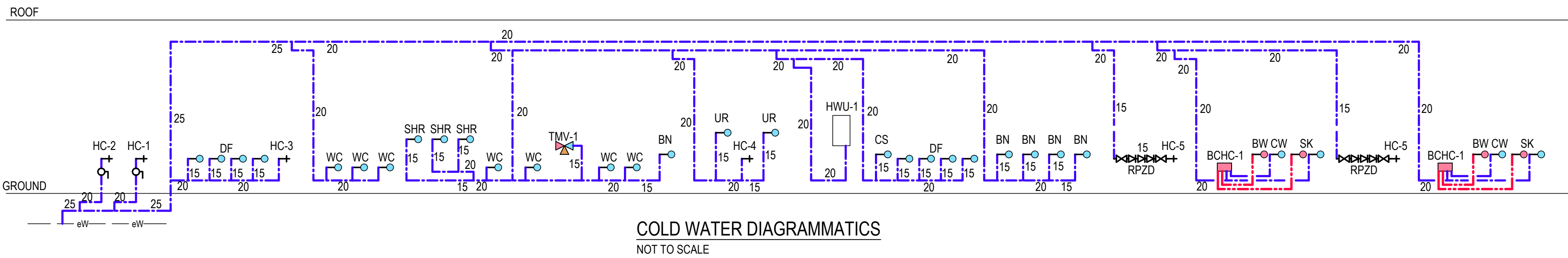
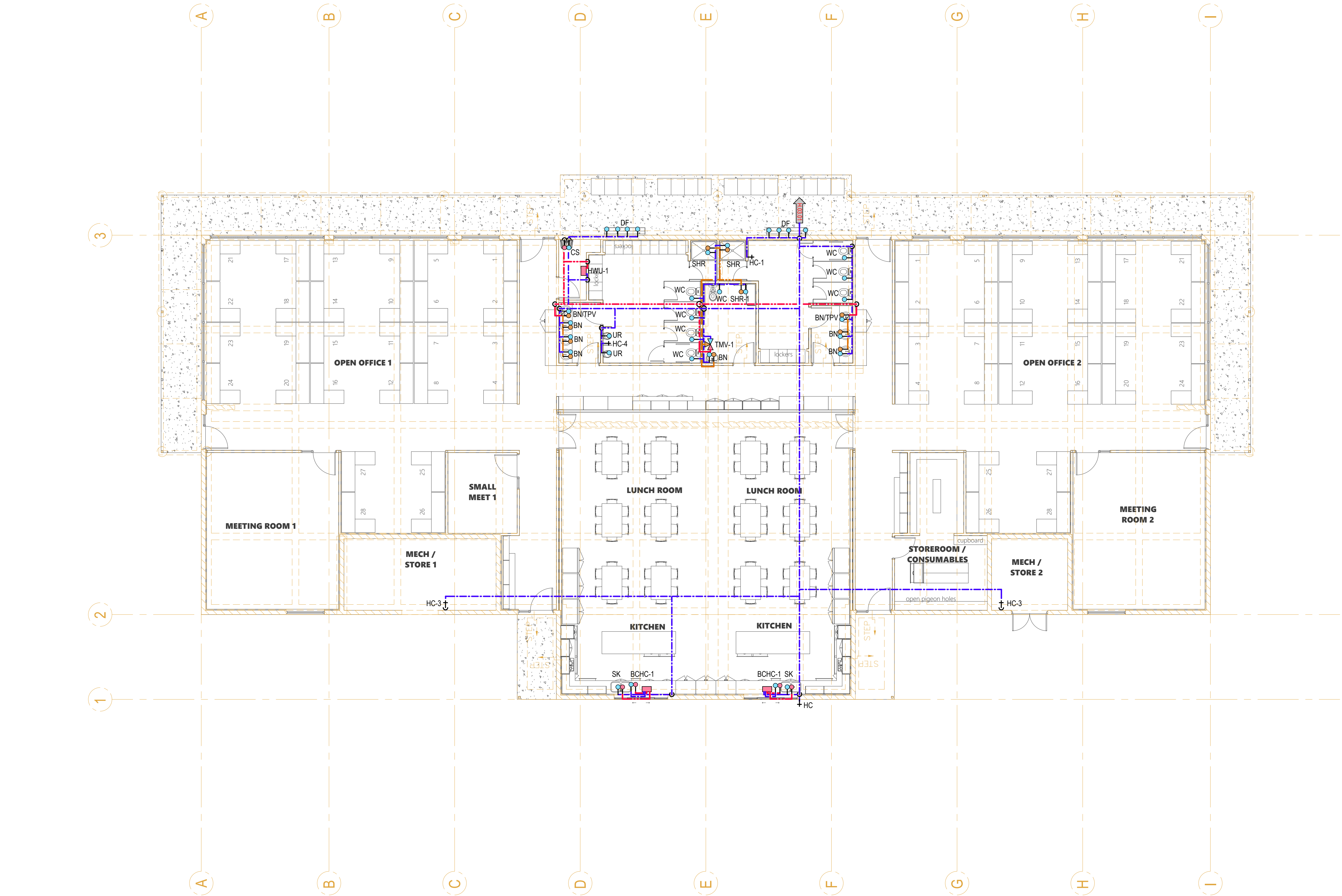
STANWELL POWER STATION
SWITCHYARD ROAD, STANWELL

TITLE:
HYDRAULIC SERVICES
GROUND FLOOR
WATER AND FIRE SERVICES
DIAGRAMMATICS

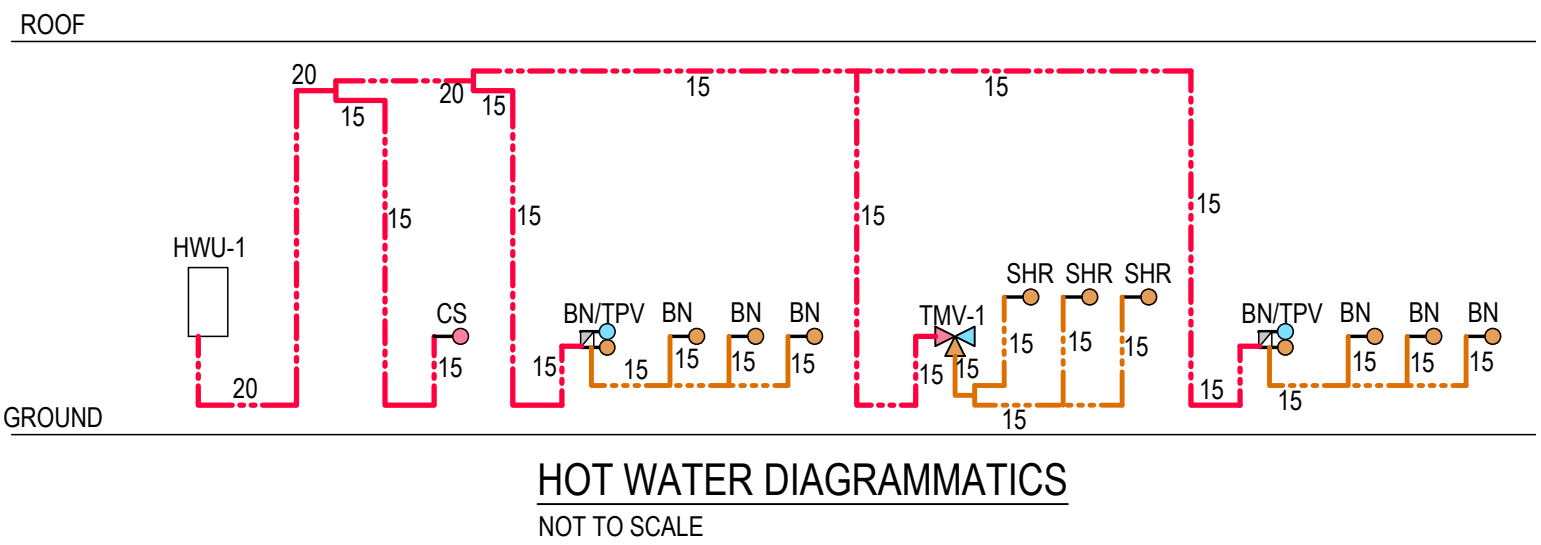
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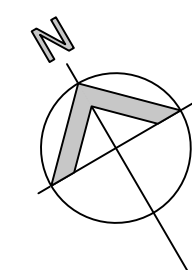
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ROOF



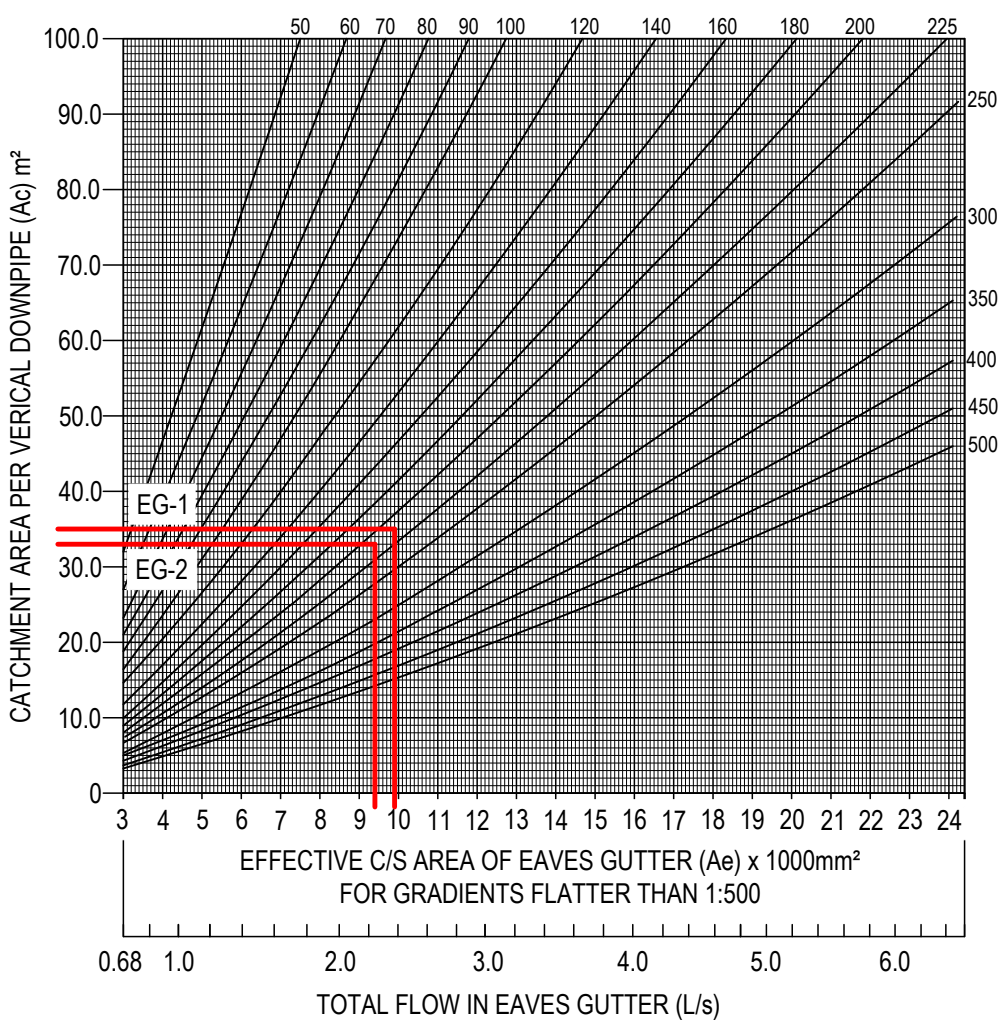
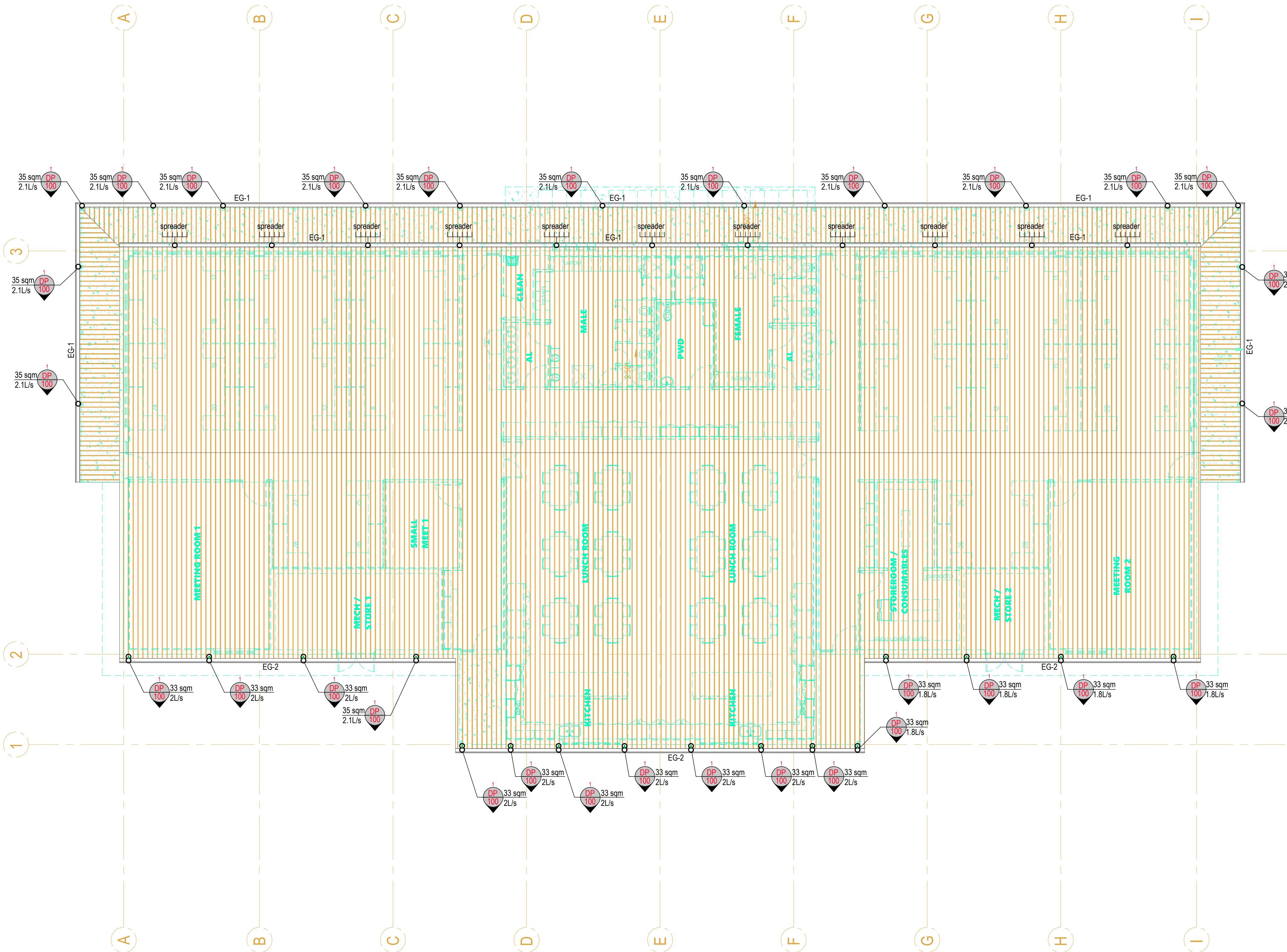
ROOF



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- NOTES:
- THIS GRAPH ASSUMES
 - AN EFFECTIVE WIDTH TO DEPTH IS RATION OF ABOUT 2:1
 - A GRADIENT IN THE DIRECTION OF FLOW IS FLATTER THAN 1:500
 - THE LEAST FAVORABLE POSITIONING OF THE DOWNSPIPE AND BENDS WITHIN THE GUTTER LENGTH
 - A CROSS SECTION OR HALF ROUND, QUAD, OGEE OR SQUARE; AND
 - THE OUTLET TO A VERTICAL DOWNSPIPE IS LOCATED CENTRALLY IN THE SOLE OF THE EAVES GUTTER
 - THE REQUIRED EAVES GUTTER DISCHARGE AREAS DO NOT ALLOW FOR LOSS OF WATERWAY DUE TO INTERNAL BRACKETS

EAVES GUTTER SCHEDULE				
gutter reference	downpipe number	rainfall intensity (mm/hr)	roof area per downpipe drop (m²)	c/s area of gutter required (mm²)
EG-1	1	217	35	9900
EG-2	1		33	9400

NOTE: EFFECTIVE C/S AREA OF FOR GUTTER GRADIENTS FLATTER THAN 1:500

EAVES GUTTER - REQUIRED SIZE OF VERTICAL DOWNSPIPE			
effective C/S area of an eaves gutter (Ae)		Internal size of vertical downpipe (mm²)	
Gradient		Cross-Section	
1:500 and steeper	Flatter than 1:500	Circular	Rectangle or square
3500	4700	65	65 x 50
4200	5600	75	65 x 50
4600	6200	75	75 x 50
4800	6400	80	75 x 50
5200	7000	80	100 x 50
5900	7900	85	100 x 50
6400	8600	90	100 x 50
6600	8900	90	75 x 70
6700	9000	100	75 x 70
8200	11000	100	100 x 75
9600	12900	125	100 x 75
12800	17100	125	100 x 100
12800	17200	150	100 x 100
16000	21500	150	125 x 100
18400	24700	150	150 x 100
19200	25800	-	150 x 100
20000	26800	-	125 x 125

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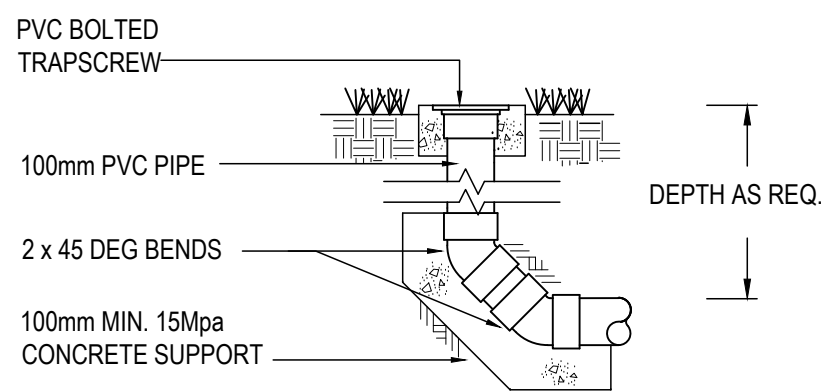
STANWELL POWER STATION
SWITCHYARD ROAD, STANWELL

TITLE:
HYDRAULIC SERVICES
ROOF PLAN
DRAINAGE

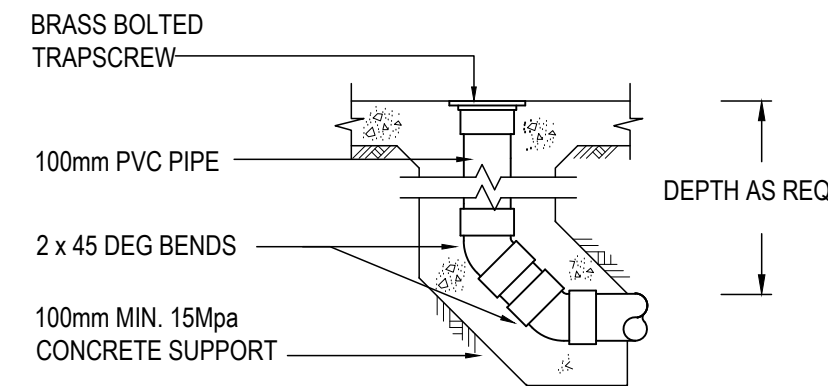
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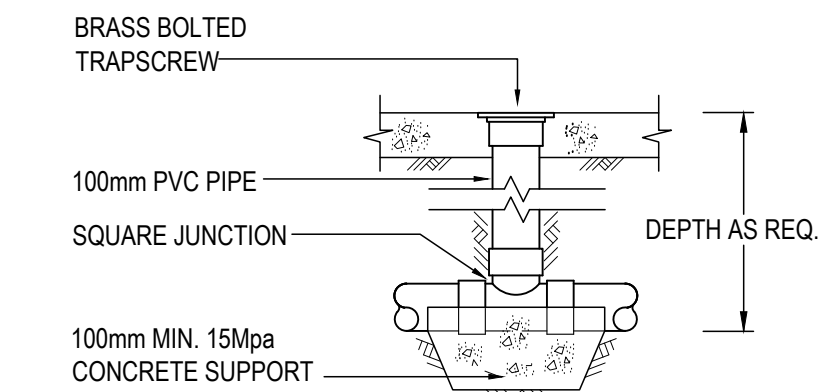
Project Number, Sheet Issue:
HS-0690-b H.04.01 01



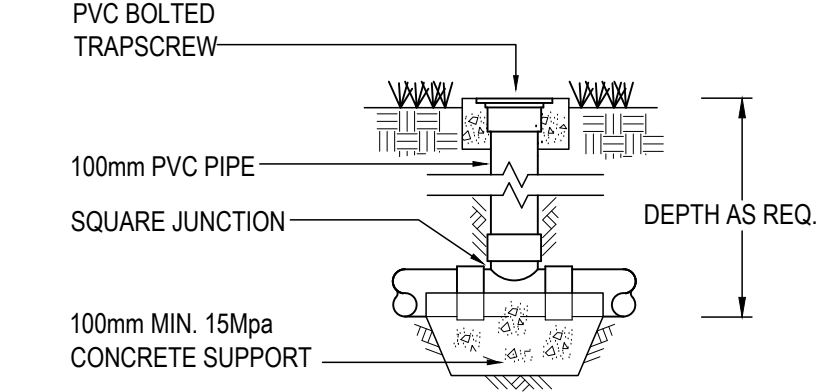
INSPECTION OPENING TO SURFACE DETAIL
(IN LANDSCAPED AREA - NON-TRAFFICABLE)



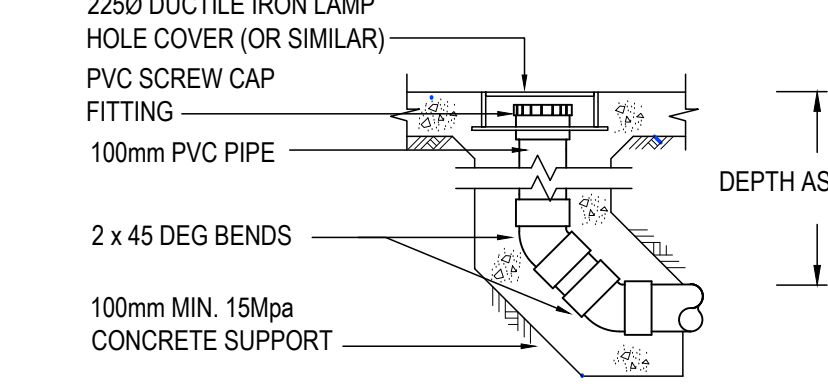
INSPECTION OPENING TO SURFACE DETAIL
(IN CONCRETE AREA - TRAFFICABLE - PEDESTRIAN)



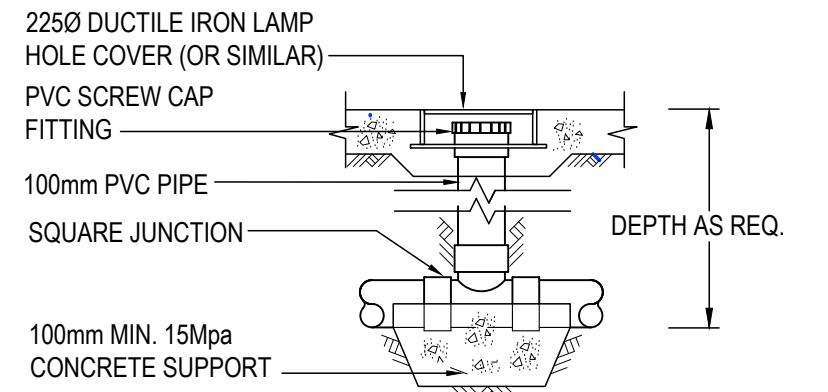
INSPECTION OPENING TO SURFACE DETAIL
(IN CONCRETE AREA - TRAFFICABLE - PEDESTRIAN)



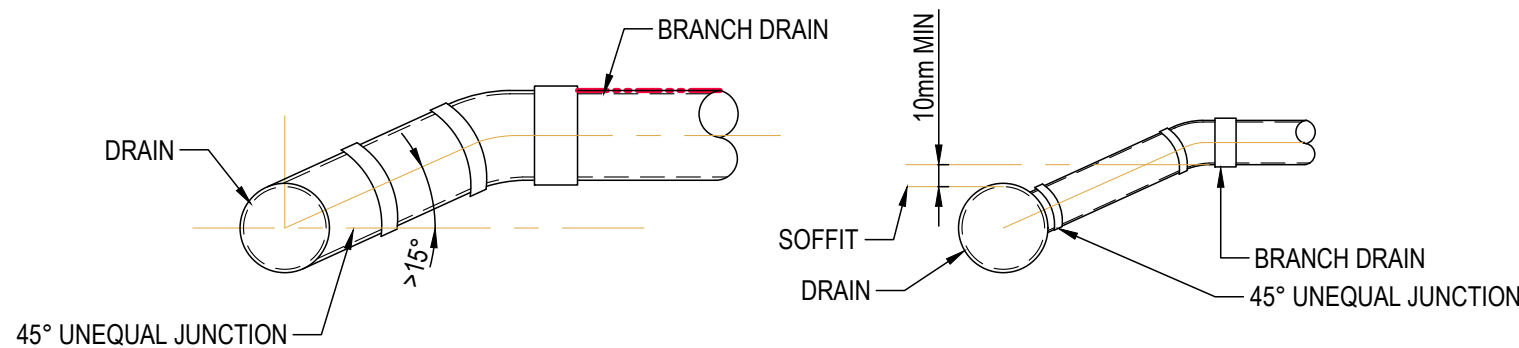
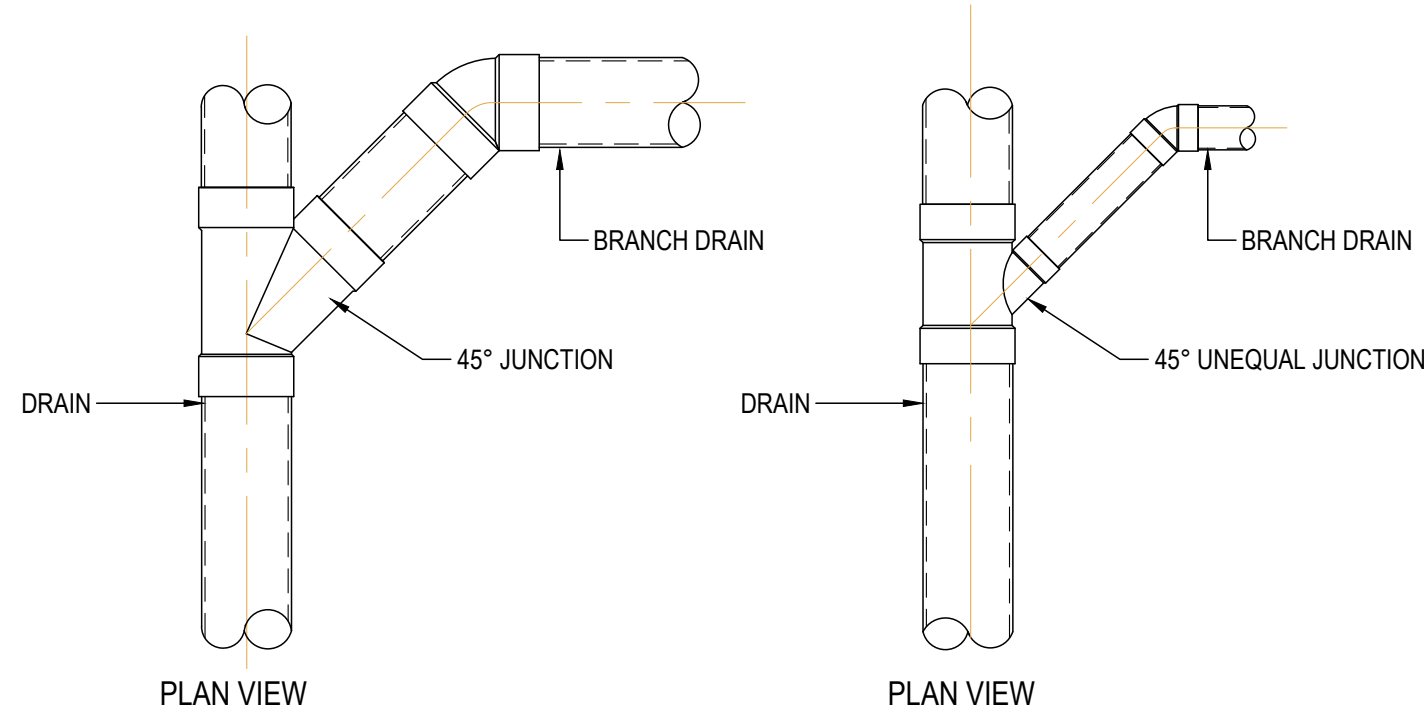
INSPECTION OPENING TO SURFACE DETAIL
(IN LANDSCAPED AREA - NON-TRAFFICABLE)



INSPECTION OPENING TO SURFACE DETAIL
(IN CONCRETE AREA - TRAFFICABLE - VEHICLE)



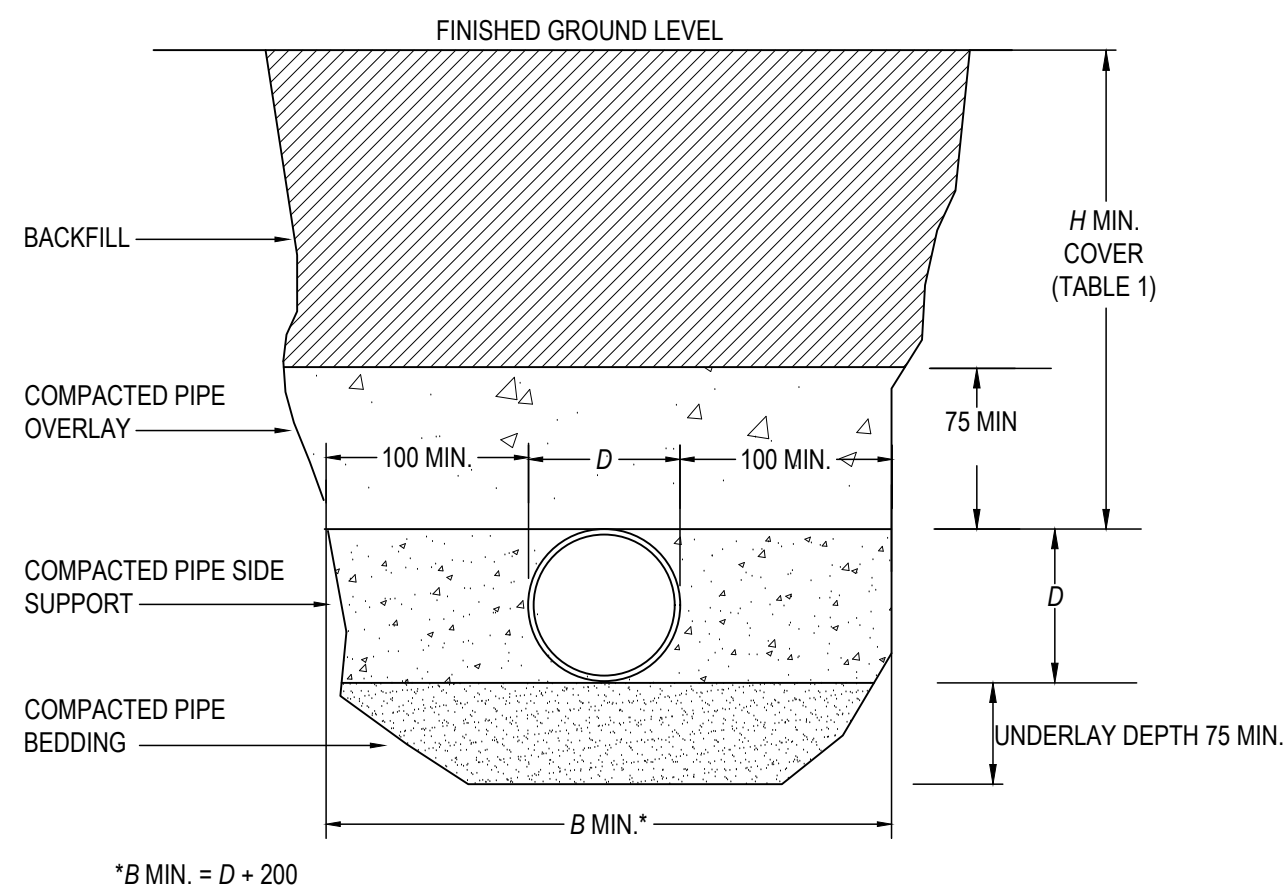
INSPECTION OPENING TO SURFACE DETAIL
(IN CONCRETE AREA - TRAFFICABLE - VEHICLE)



(a) JUNCTION AT GRADE
(b) UNEQUAL JUNCTION AT GRADE

GRADED BRANCH DRAIN ENTERING ANOTHER DRAIN DETAIL

NOT TO SCALE
NOTE: A JUNCTION IS NOT REQUIRED TO BE INCLINED 15° ABOVE THE HORIZONTAL WHERE THE INSTALLATION IS;
• FOR A JUNCTION INSTALLED AS PARTY OF A REPAIR OR EXTENSION TO AN EXISTING INSTALLATION
• WHERE THERE ARE NO WC PANS CONNECTED UPSTREAM OF A JUNCTION ON EITHER A DN100 BRANCH DRAIN OR DN100 MAIN DRAIN
• WHERE A JUNCTION IS NOT A DN100 x DN100 JUNCTION, E.G. A DN100 x DN65 OR DN150 x DN100 JUNCTION

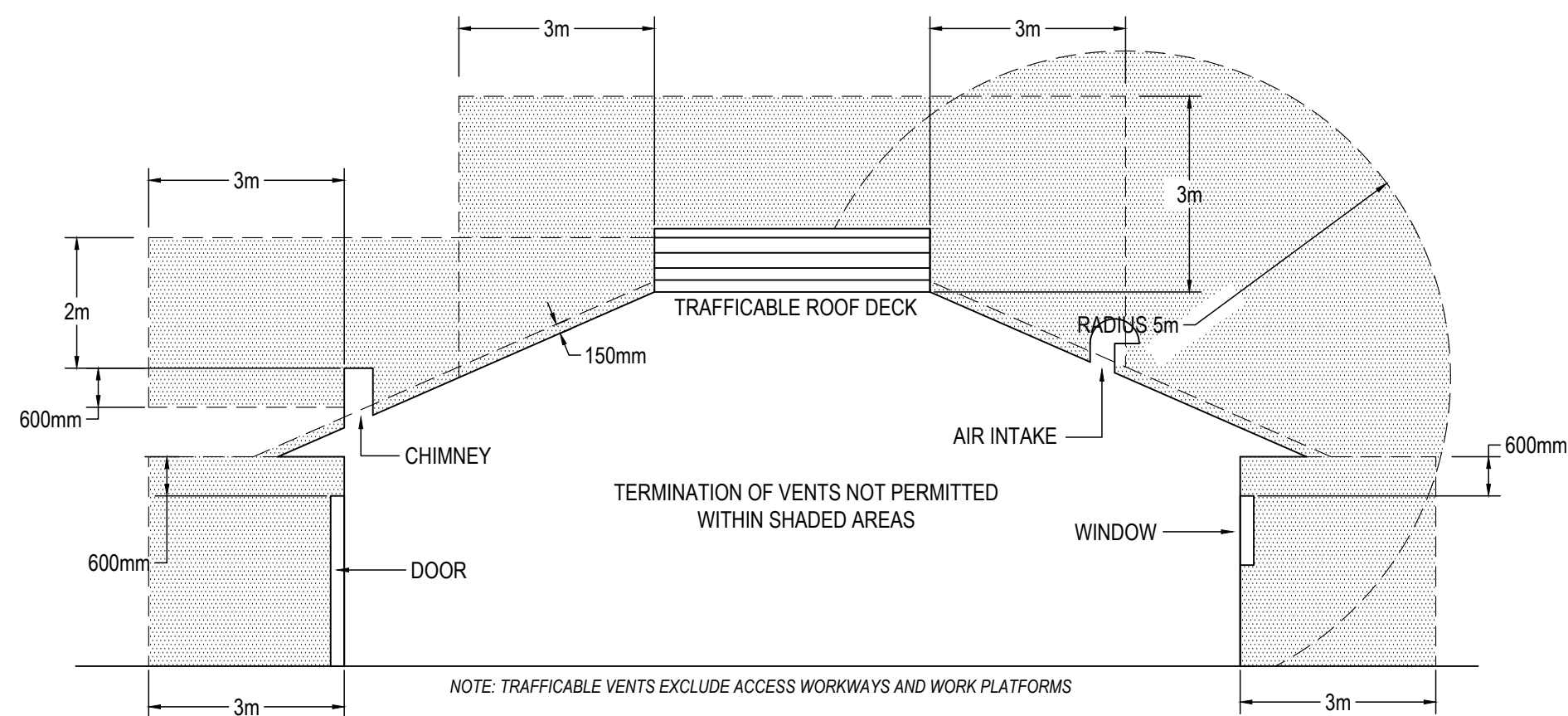


TYPICAL BEDDING OF SANITARY DRAINS DETAIL

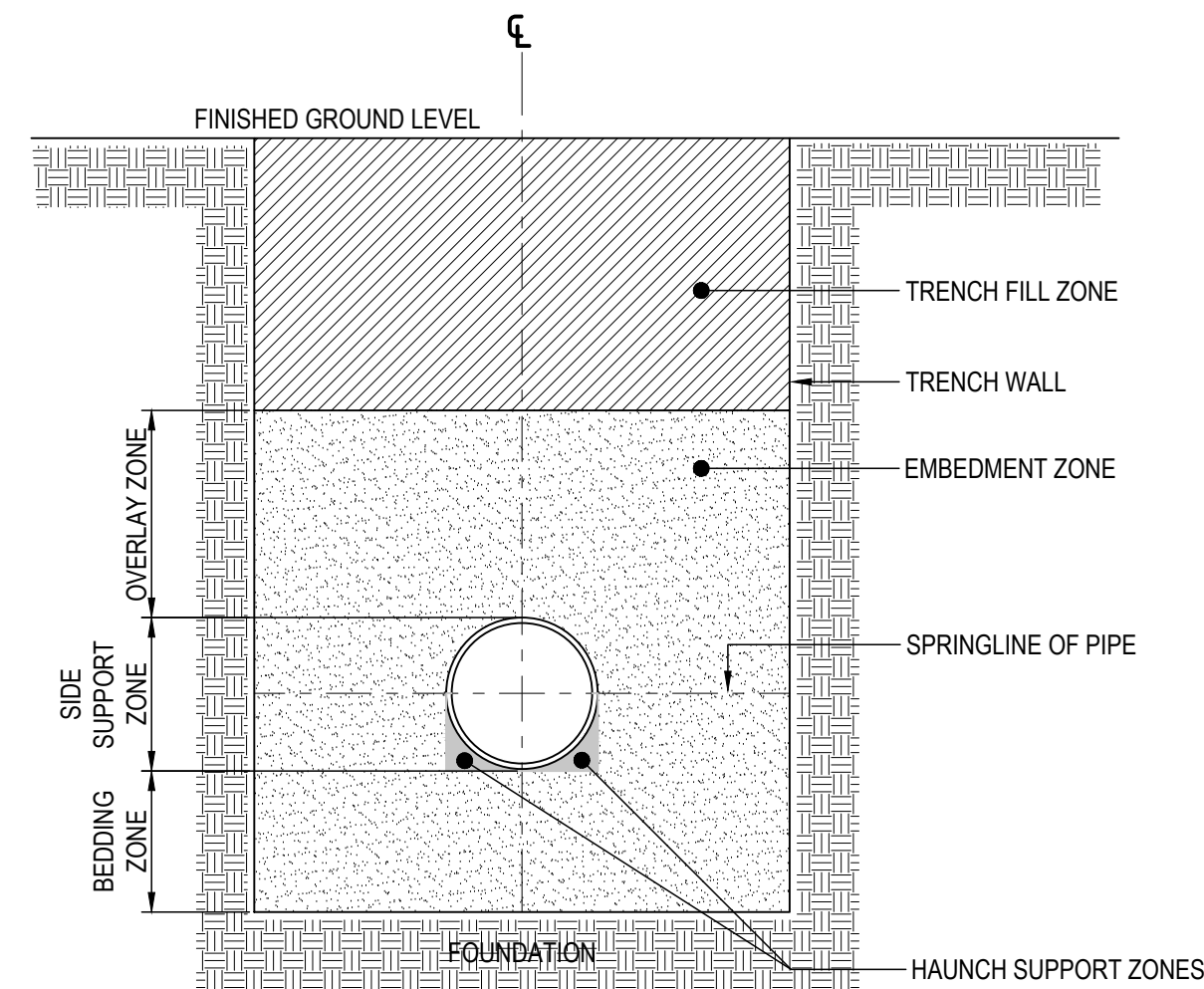
NOT TO SCALE

Table 1 - Minimum depth of cover for buried sanitary drain pipes		
Location	minimum cover (mm)	
	cast & ductile iron	other material
Subject to vehicular traffic	300	500
all other locations	NIL	300*

* alternatively drains covered by at least 50mm of overlay and then paved with-
1. 100mm thick reinforced concrete where subject to heavy vehicular traffic
2. 75mm brick or concrete paving where subject to light vehicular traffic; or
3. 50mm brick or concrete paving where not subject to vehicular traffic



TERMINATION OF VENT DETAIL
NTS



TYPICAL BEDDING OF STORMWATER DRAINS DETAIL

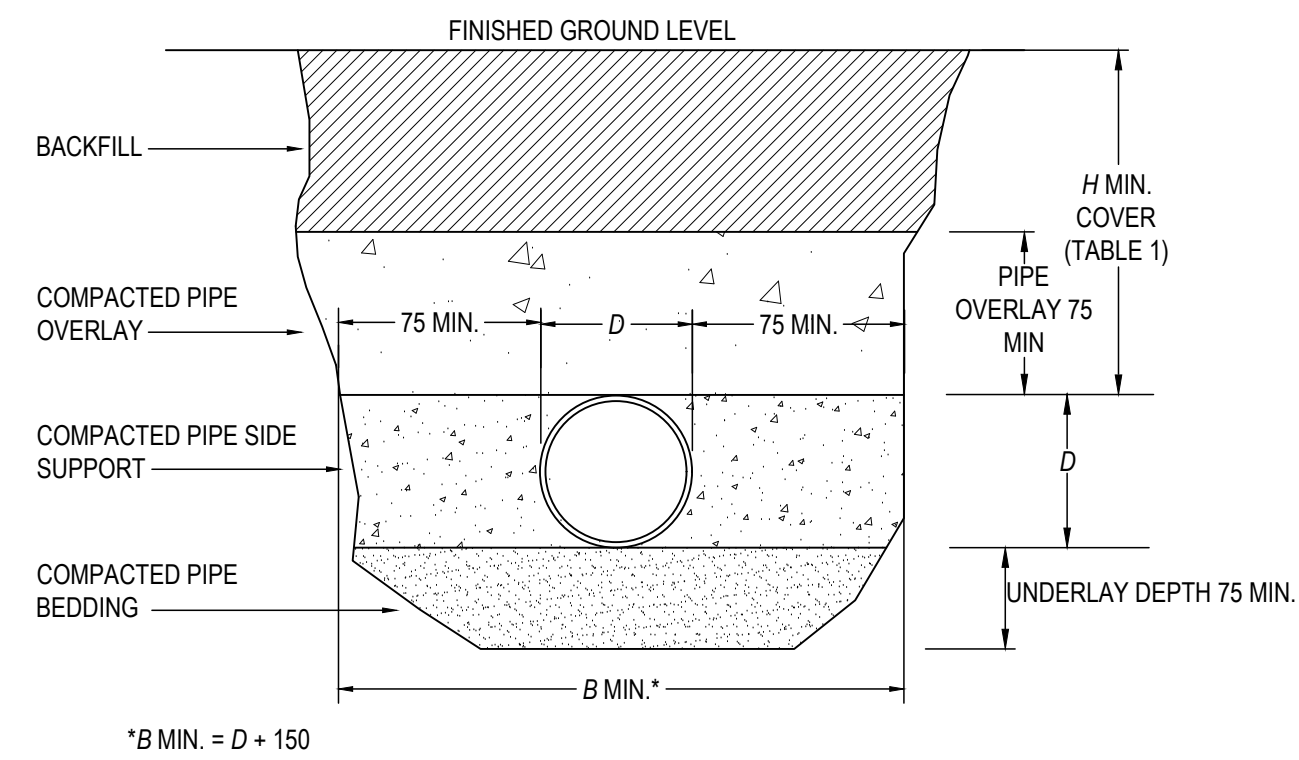
NOT TO SCALE

Notes:
1. Trench widths measured at the top of the pipes, between the faces of either the unsupported trench walls or the inside face of the sheeting of the trench support system, shall be not less than the widths specified in -
(a) AS/NZS 2041.1 and AS/NZS 2041.2 for corrugated metal pipes;
(b) AS/NZS 2566.2 for flexible pipes and fittings;
(c) AS/NZS 3725 for FRC and reinforced concrete pipes; and
(d) AS 4060 for vitrified clay and ceramic pipes and fittings.

Table 1 - Minimum depth of cover for buried stormwater drain pipes

Location	minimum cover (mm)	
	ductile iron, galvanised steel	plastics
1. Subject to vehicular traffic		
(a) without pavement		
(i) for single dwellings	100	100
(ii) for other than single dwellings	100	300
(b) with pavement of brick or unreinforced concrete	100*	100
2. subject to vehicular load		
(a) other than roads		
(i) without pavement	300	450
(ii) with pavement of -		
- reinforced concrete for heavy vehicle loading; or	NIL*	100*
- brick or unreinforced concrete for light vehicular loading	NIL*	75*
(b) Roads		
(i) sealed; or	600	600
(ii) unsealed	600	750
3. Subject to construction equipment loading or in embankment conditions	600	750
4. Land zone for agricultural use	600	600

* below the underside of pavement

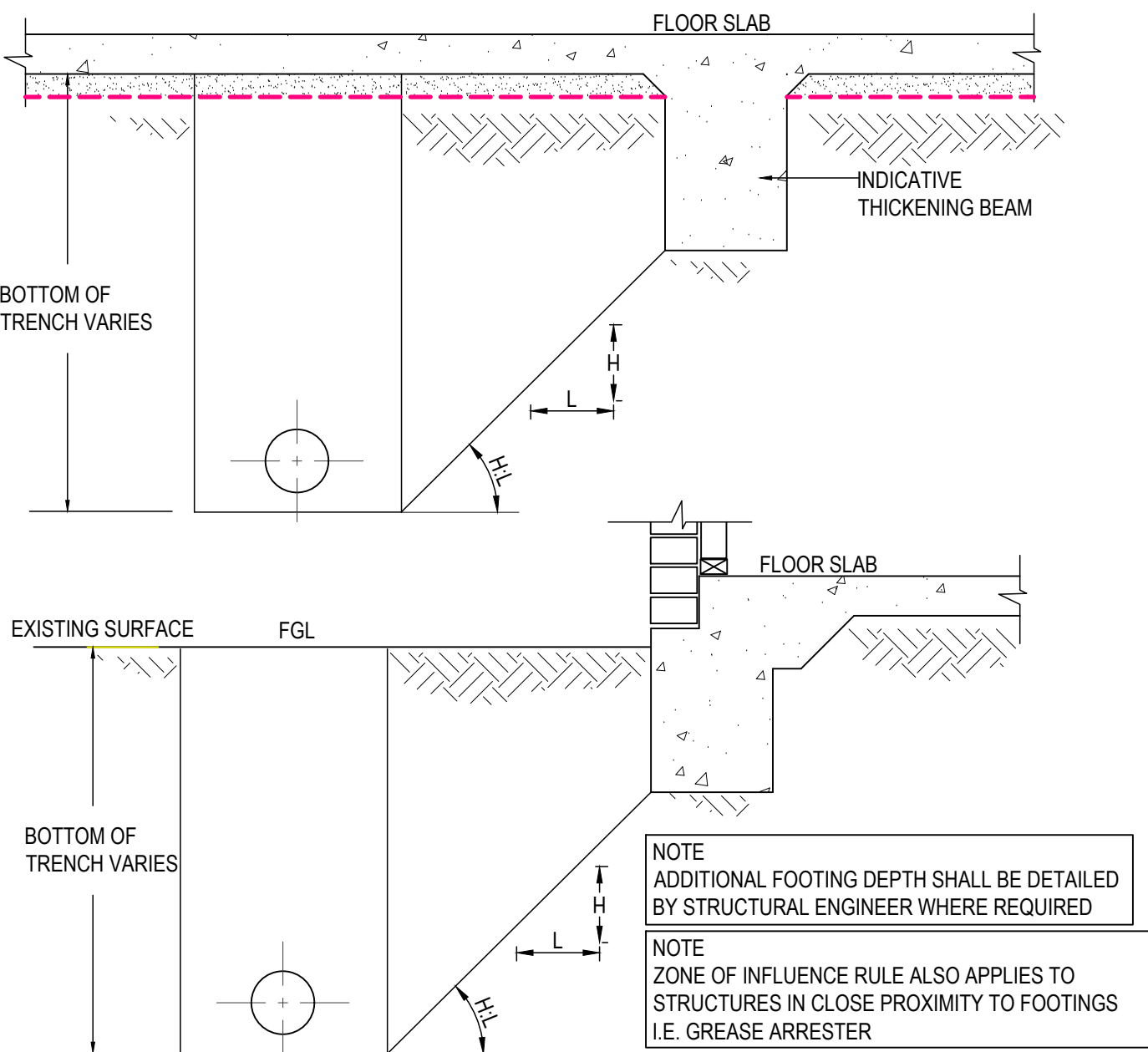


*B MIN. = D + 150

TYPICAL BEDDING OF WATER SUPPLY PIPES DETAIL

NOT TO SCALE

Table 1 - Minimum depth of cover for buried water pipes	
Loading Conditions	minimum cover (mm)
Under slab and footings	75
not subject to vehicular loading (excluding fire services)	300
fire services not subject to vehicular loading	600
subject to vehicular loading:	
(a) no carriageway	450
(b) sealed carriageway	600
(c) unsealed carriageway	750
pipes in embankments or subject to construction equipment loads	750



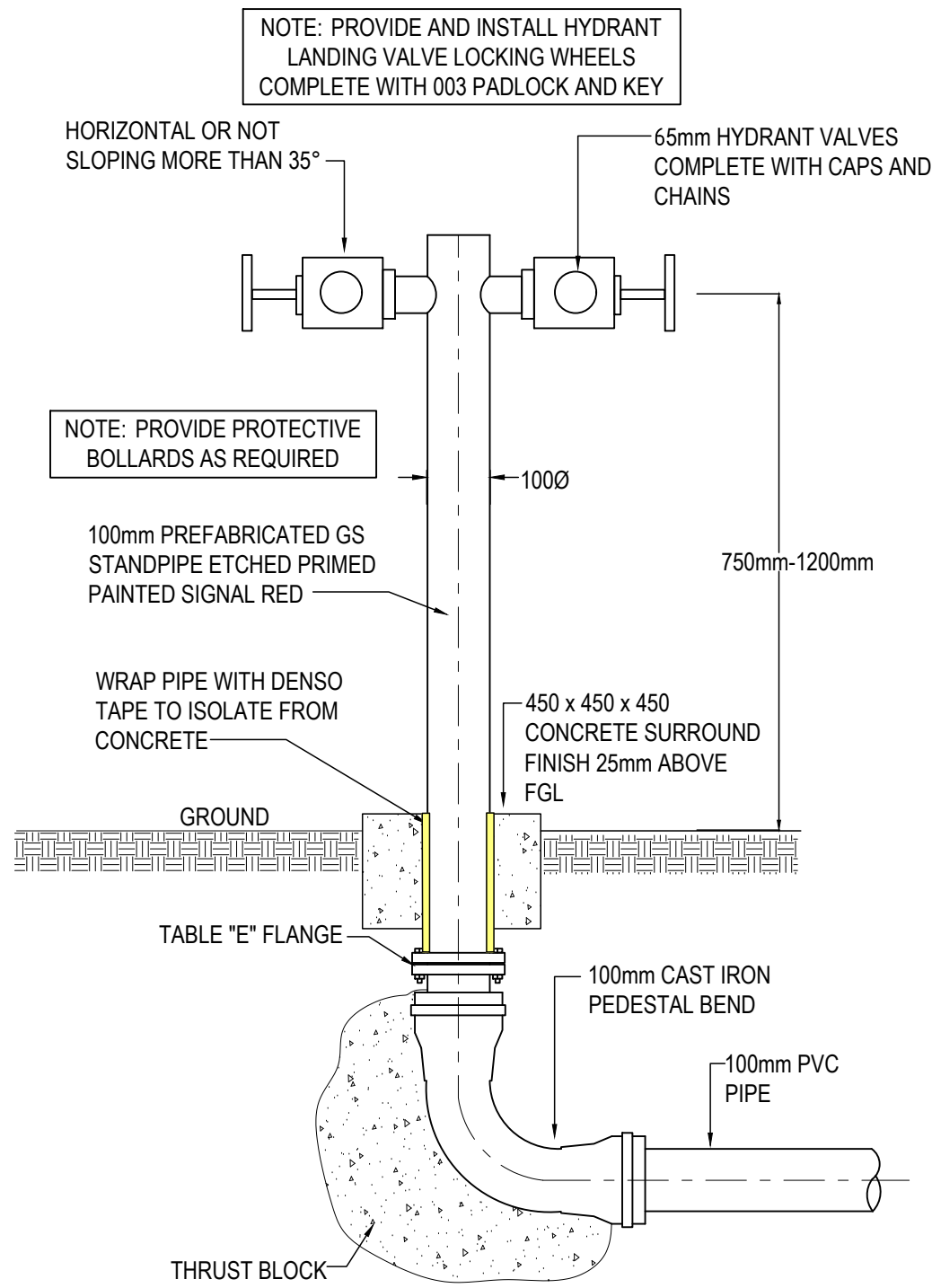
Soil Type	Slope H:L	
	Compacted Fill	Undisturbed Ground
Stable Rock*	2:3	8:1
Sand*	1:2	1:2
Silt**	1:4	1:4
Firm Clay	1:2	1:1
Soft Clay	Not Suitable	2:3
Soft Soils**	Not Suitable	Not Suitable

* Most sand & rock sites with little or no ground movement from moisture changes

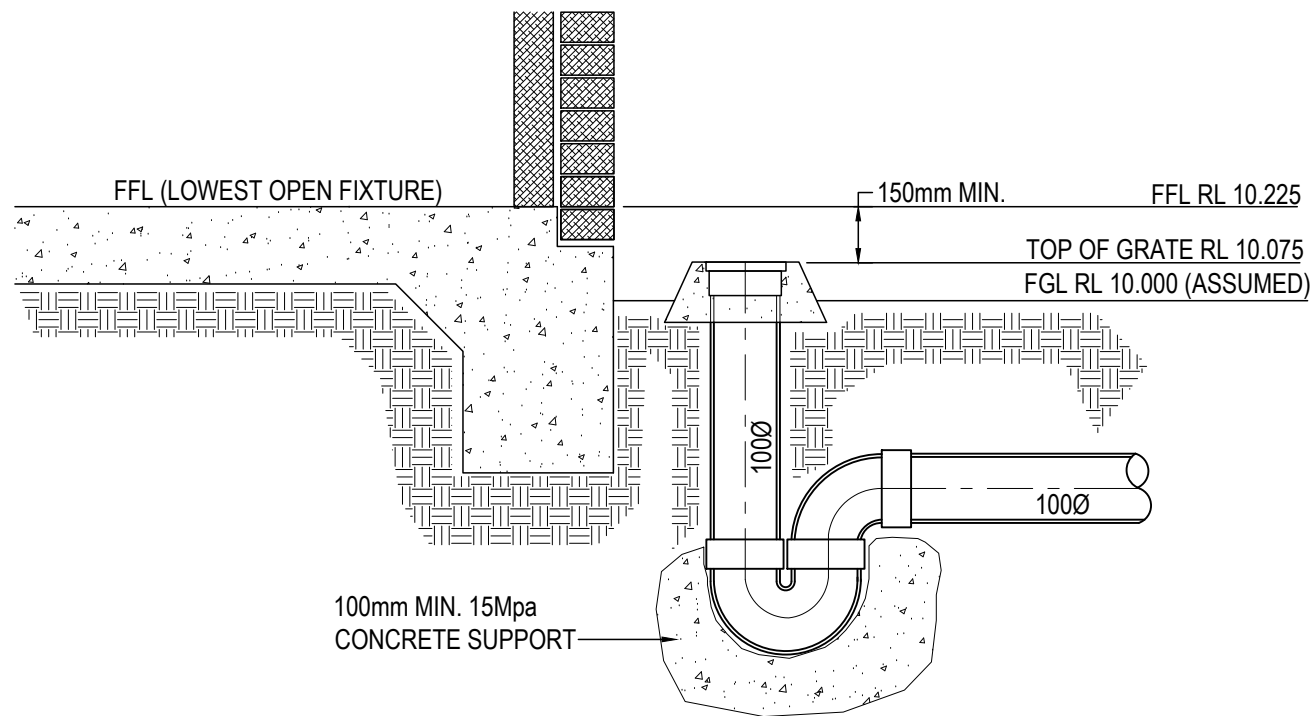
** Sites include soft soils such as soft clay or silt or loose sand, landslide, mine subsidence, collapsing soils, soils subject to erosion, reactive sites subject to abnormal moisture conditions or sites that cannot be classified otherwise

EXCAVATION NEAR FOOTING DETAIL

NOT TO SCALE

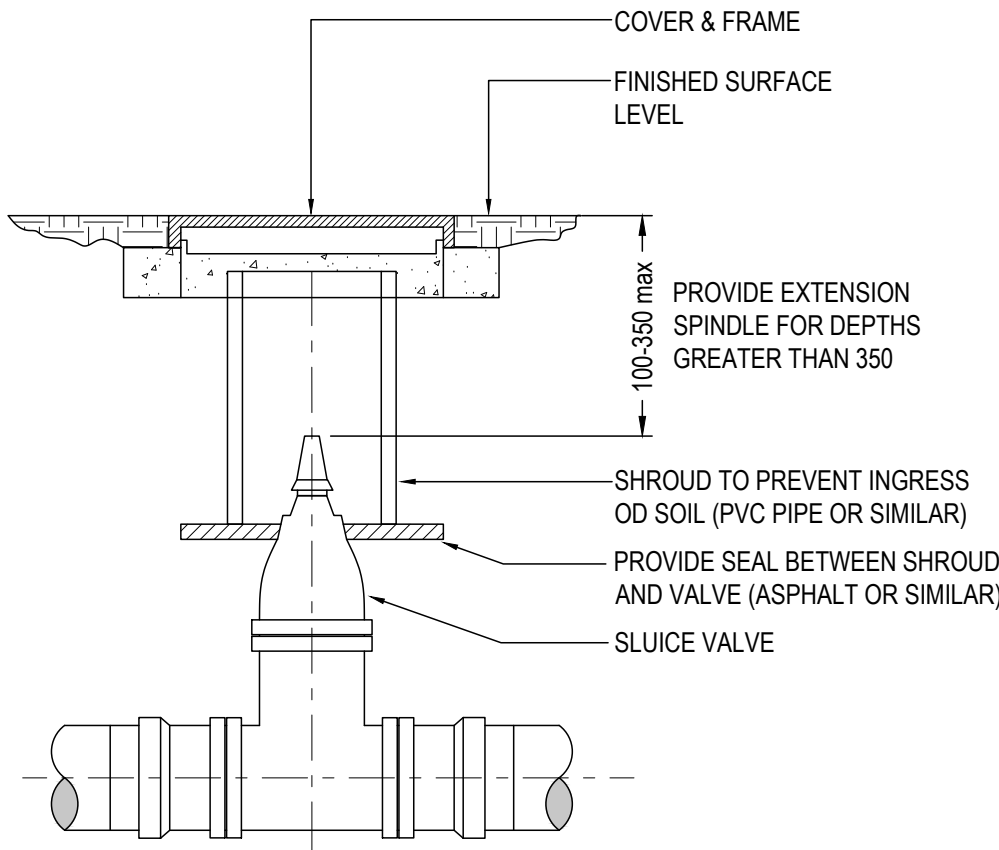


DUAL PILLAR HYDRANT DETAIL
NOT TO SCALE

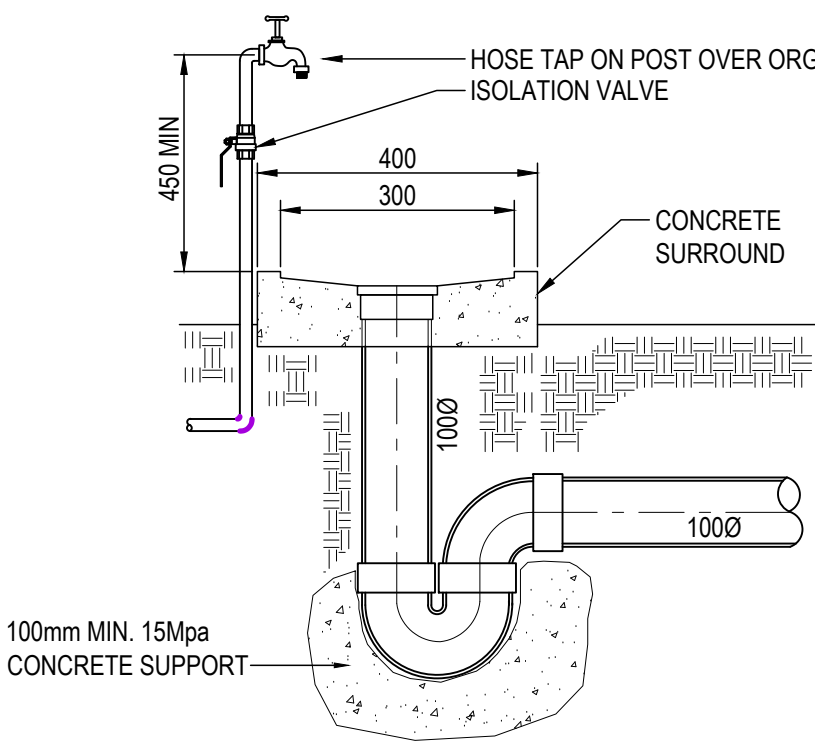


INDICATIVE ORG LEVEL DETAILS
NOT TO SCALE

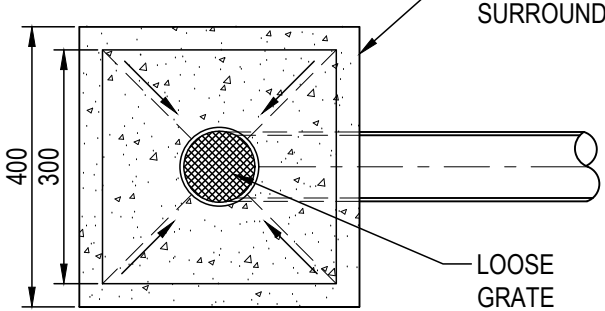
POINT OF MEASUREMENT ON FIXTURES FOR HEIGHT ABOVE OVERFLOW GULLY	
FIXTURE	POINT OF MEASUREMENT
SOIL FIXTURE WITH INTEGRAL TRAP	TOP SURFACE OF THE WATER SEAL
FLOOR WASTE GULLY OR SHOWER	TOP SURFACE LEVEL OF THE GRATE
SOIL FIXTURE LOCATED IN AN OUTBUILDING OR ROOM, THE FLOW OF WHICH IS GRADED TO AN EXTERNAL DOORWAY	OVERFLOW RIM OF FIXTURE
OTHER FIXTURES (INCLUDING GREYWATER DIVERSION DEVICES)	TOP SURFACE LEVEL OF FIXTURE OUTLET



TYPICAL BELOW GROUND SLUICE VALVE DETAIL
NTS



SECTION



PLAN

HOSE TAP OVER ORG DETAILS
NOT TO SCALE

- NOTE
- FIX SIGN TO ADJACENT STRUCTURE WITH CORROSION RESISTANT FASTENERS
 - THE SIGN SHALL BE COLOURED IN ACCORDANCE WITH AS 1319



TYPICAL NON-POTABLE WATER SUPPLY SIGN
NOT TO SCALE

02	REVISION	09.03.2026
01	ISSUED FOR TENDER	24.02.2026
ISSUE:	DESCRIPTION:	DATE:
HTD	HYDRO-TECH DESIGNS PTY LTD HYDRAULIC AND FIRE PROTECTION CONSULTANTS Tel: 0438 433 544 Email: rjh@bigpond.net.au	

CLIENT:



BUILDING DESIGNER



PROJECT:
NEW OFFICE BUILDING

STANWELL POWER STATION
SWITCHYARD ROAD, STANWELL

TITLE:
HYDRAULIC SERVICES
DETAILS

Datum:	Status: TENDER	
Scale:	AS NOTED	Date: 19.11.2025
Project Number:	Sheet	Issue:
HS-0690-b	H.05.02	02